



Block-3

Digital Journalism

THE PEOPLE'S
UNIVERSITY

BLOCK 3: DIGITAL JOURNALISM

Introduction

Unit 11: Issues with Digital Inequality

Our exploration begins with “Issues with Digital Inequality,” delving into the conceptual development of digital inequality in contemporary society. We examine the technological approach to digital inequality, exploring the idea of digital capital and the nuanced dynamics of inclusion and exclusion. This unit serves as a foundation for understanding the socio-economic disparities amplified by digital advancements.

Unit 12: Digital Media and Environmental Sustainability

Turning our attention to the intersection of digital media and environmental sustainability, “Digital Media and Environmental Sustainability” scrutinizes the impact of Information and Communication Technologies (ICT) on lifestyle choices. We delve into the key pillars of sustainable development, explore the concept of Green ICT, and assess the roles of governments and corporations in fostering environmental responsibility through digital journalism.

Unit 13: Implementation of ICT for Development

In the final leg of our journey, “Implementation of ICT for Development” takes a practical approach to key concepts, presenting case studies that highlight the transformative potential of ICT. We navigate through challenges and ethical considerations, exploring the intricate landscape of utilizing digital tools for societal development. This unit empowers you to critically analyze the role of digital journalism in addressing contemporary issues and shaping a more equitable and sustainable future.

As we conclude this exploration into “Digital Journalism,” we invite you to reflect on the profound impact of digital technologies on journalism, society, and the environment. Armed with insights into digital inequality, environmental sustainability, and the implementation of ICT for development, you are poised to navigate the complexities of the digital age as a responsible and informed journalist. Embrace the transformative power of digital media while being mindful of the challenges and ethical considerations that accompany this dynamic landscape. Through this block, we aim to empower you to contribute meaningfully to the evolving field of digital journalism.

UNIT 11 ISSUES OF DIGITAL INEQUALITY

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11.0 INTRODUCTION

The proliferation of information and communications technology (ICT) has given prominence to the digital divide discourse, which has become a subject of global concern. With the rise of new ICTs in the second half of the 1990s, the notion of the digital divide emerged in the policy context to address unequal access to digital technologies. All countries do not have the same ICT penetration or growth rate. Developing countries, particularly in Africa, Asia, and Latin America, often face significant challenges regarding access to digital infrastructure, affordability of digital devices, and availability of

digital skills and knowledge. Although it has become hard to imagine life without the Internet and other forms of ICT, only some have the same access to devices such as phones, tablets, laptops, and desktop computers. Thus, the discussion over access to digital technologies has given rise to a debate over their usage from the standpoint of equality.

Simply put, digital inequality results from unequal access, skill, and use of the Internet and other ICTs. The conceptual framework for understanding digital inequality deals with the causes and consequences of digital inequality concerning ICTs, considering those as empowerment tools. Several disciplines contribute elements to the framework of digital inequality. Sociology emphasises social inequality regarding access to resources, control over different types of capital, and social involvement. Psychology examines problems as well as attitudes and reasons for using technology. Economics emphasises the spread of relevant innovations. Education deals with digital literacy, skills, and competence. Above all, communication studies deal with the tangents of mediated communication by constructing a connection between elements. Therefore, addressing the concept through the lens of communication studies requires a multi-faceted approach that involves sociological and technological perspectives intertwined with psychological, educational, and economic dimensions.

This Unit describes conceptual and theoretical patterns for identifying and understanding digital inequality using socio- and techno-centric accounts. Although only a small amount of theoretical study has been done so far on digital inequality, the Unit throws light on traditional as well as emerging concepts and frameworks that revolve around the rise of the digital divide, the manifestation of digital inequality, multiple concepts related to its components, and new forms of digital inequality.

11.1 LEARNING OUTCOMES

After reading this chapter, you will be able to understand:

- multidimensionality of digital inequality;
- nuanced conceptual and theoretical approaches in the field of digital inequality;
- digital Inequality from a Sociological Perspective; and
- technological approach to digital inequality.

11.2 DIGITAL INEQUALITY: A SOCIAL PROBLEM

Neither the essences of individuals nor the essences of specific collectives or systems (such as capitalism or patriarchy) serve as the starting point for this idea of inequality (van Dijk, 2005). Instead, the idea of inequality is based on the connections, relationships, interactions, and transactions that exist

between people. Therefore, digital inequality is a societal rather than a technical issue.

Access to digital technologies is necessary for participation in many aspects of life, including education, employment, healthcare, and social interaction. Individuals with limited access to digital technologies may find it difficult to fill out online job applications, communicate with potential employers, access online training resources, etc. Digital inequality is a social issue because it creates and expands social inequalities in education, employment, and information access. Those with limited access to digital technologies and skills may experience significant disadvantages in these areas, affecting their social and economic well-being.

Digital inequality can exacerbate other forms of social inequality, such as those based on race, gender, and socioeconomic standing. Those already marginalised in these areas may experience even more difficulty gaining access to and utilising digital technologies, which can exacerbate existing social disparities. As an example, in India, the digital gender gap encourages gender inequity. With internet connectivity, women may be included in employment opportunities, educational opportunities, and social networks, restricting their social mobility and economic desires. Digital inequality is a social concern because it creates and reinforces social inequalities, which can have long-term repercussions for individuals and society.

11.3 UNDERSTANDING THE CONCEPTUAL DEVELOPMENT

The term "digital divide" was not coined by any theorist; instead, it was first used in an official publication by the National Telecommunications and Information Administration (NTIA, 1999) in North America. In the earliest accounts of the phenomenon, the "digital divide" meant a disparity in the likelihood of having access to information and communication technologies (ICT) based on demographic characteristics such as race, gender, age, socioeconomic status, level of education, and the composition of the household. As the prevalence of computers and the Internet increased in developed nations, the discourse began to evolve, and more intricate conceptualisations were devised. In a short period, research on the digital divide has gained importance as an academic field; as a result, the concept of the "digital divide" has become more comprehensive with the emergence of "digital inequality".

11.3.1 Shift From Divide to Inequality

The term "digital divide" originally referred to the disparity between people with access to modern technologies and those without access (Van Dijk, 2005). This definition of the digital divide has been observed in terms of physical access to telephones, personal computers, cellular devices, etc. Later on, the definition became discordant with understanding the multidimensionality of communication beyond physical access to

technologies. Moreover, it has become clear to social scientists that the digital divide is not related to a single type of binary gap but is intertwined with a range of social, economic, and technological issues. As soon as social scientists observed the nuances of digital access's social causes and consequences, the term got another social dimension called "inequality". According to this concept, the differences in access and usage of digital technologies are directly or indirectly related to social, cultural, political, and economic inequalities in society.

11.3.2 Dimensions of Digital Inequality

The digital divide laid the foundation for the concept of digital inequality. To understand the concept of digital inequality, one needs to understand the levels of the digital divide. Three levels of the digital divide correspond to multiple dimensions of digital inequality.

Table 1. The three levels of the digital divide

Levels	Deals With
First level	Binary access to ICTs (have or have not)
Second level	Digital skills
Third level	The tangible outcome of digital usage

Source: Author's compilation

The first level: The first level of the digital divide describes the unequal distribution of access to digital technologies. Low-income persons, those living in rural regions, people with impairments, and members of racial and ethnic minorities are disproportionately impacted by this divide.

Second level: The second level of the digital divide refers to the unequal distribution of digital skills and competence among people with access to digital technologies. This divide affects those with access to digital technologies but lacking the skills and competence to utilise them effectively. Individuals from low-income or marginalised communities may lack access to education or training in digital skills, resulting in the second level of the digital divide.

Third level: Access to digital devices and acquiring digital skills often fail to predict evenly distributed digital participation (meaningfully participating in the digital sphere). The third level of divide refers to the disparities in abilities to use digital technologies to achieve tangible outcomes of technology usage, such as online learning, remote work, e-commerce, and civic engagement.

11.3.3 Concept of Digital Inequality

The term "digital inequality" reflects a broader conception of the digital divide and incorporates a variety of factors beyond access to technology,

including skills, usage patterns, and stakeholders involved in the acquisition of empowerment through technology. It acknowledges that more than merely providing access (binary access to digital technologies) to technology is required to ensure equal benefits from the opportunities it provides. Unequal access to economic, social, cultural, and personal resources impacts digital engagement (Helsper, 2012). Inequality may limit or boost citizens' social, economic, political, personal, and cultural capital, affecting their access to essential knowledge and ability to engage in society (Van Dijk, 2005). A range of determinants contribute to disparities in access to and use of digital technologies.

Table 2. Determinants of digital inequality

Socio-demographic	Age, gender, marital status, residency, living area
Economic	Income, employment status, employment type, occupational status, educational level
Social	Online social interaction, social networking, types of social activities
Cultural	Religion, ethnicity, internet use, language
Personal	Type of online activity language skills, English skills, cognitive function
Material	Internet access, access locations, number of electronic devices
Motivational	Attitude towards ICTs, internet motivation, frequency of internet use, time spent online

Source: Adapted from Scheerder, Deursen, & Dijk (2017)

11.4 DIGITAL INEQUALITY AND CONTEMPORARY SOCIETY

The information and network society context best describes high-tech cultures of particular interest that cater to the digital environment. Scholars, including Manuel Castells and Jan van Dijk, used information and/or network societies to discuss disparities in ICTs. Although developed civilisations often display more information and network society features, discussing these concepts must be addressed in pursuing digital inequality, even in other sections of the world.

11.4.1 Inequality in the Information Society

The idea of an information society broadly describes cultures in which information increasingly serves as both the primary input and output for all processes. Manuel Castells (2011) says that an information society is a type

of social organisation in which information creation, processing, and transmission become the primary sources of productivity and power. Knowledge, information, and data continually expand and advance quickly in the information society. Inequality in the information society appears at the most fundamental level of information need, which leads to a widening gap in the capacity for association and knowledge transfer and in the capacity to judge the quality of information and make other related decisions. In modern times, there is a considerable disparity in the distribution of information abilities among different groups of people.

11.4.2 Inequality in the Network Society

The concept of network society is not synonymous with the concept of information society; instead, it is an extension of it. The notion of an information society emphasises the changing nature of activities and processes in modern developed nations. According to Castells (2011), the network society is an informational society in which networks serve as the core organisational structure and pervade all areas. Van Dijk (2006) characterised the network society as an information society shaped by a nervous system of social and media networks. Networks are formed when specific actors select others to join them. As a result, individuals or organisations are either included or removed. Being unable to connect to these networks entails complete exclusion and digital marginalisation.

Check Your Progress: 1

Note: 1) Use the space below for your answers.
2) Compare your answers with those given at the end of this unit.

1. Explain the concept of “digital inequality”.

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2. What are the levels of digital inequality?

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3. Mention any three essential determinants of digital inequality.

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4. How does inequality affect the information society?

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11.5 TECHNOLOGICAL APPROACH TO DIGITAL INEQUALITIES

Research-based theories and models help us make sense of complex ideas. Digital inequality, an interconnected and complex phenomenon, can be observed through various theoretical frameworks. Over the past decades, several theories regarding the unequal distribution of ICTs have emerged from studying their introduction, diffusion, and use. Theoretical frameworks for comprehending inequalities regarding how people adopt and integrate new technologies to be discussed include the Diffusion of Innovation Theory, the Unified Theory of Acceptance and Use of Technology (UTAUT), and the Resources and Appropriation Theory (RAT).

11.5.1 Diffusion of Innovation Theory

The diffusion of innovation theory was developed by E.M. Rogers in 1962 to describe how a community of potential customers adopts and spreads an innovation. This theory constructs the pattern by which new ideas, practises or products spread through a population. In this process of innovation and diffusion, diffusion occurs when individuals go through the steps of being aware of the need for an invention, deciding to adopt or reject the innovation, trying out the innovation for the first time, and then using the innovation regularly. Rogers divided adopters into five groups according to how long it took them to decide to adopt. The categories are innovators, early adopters, early majority, late majority, and laggards.

Innovators: People willing to take risks are the first to try new ideas.

Early adopters: Those eager to test the latest innovations and determine their application for everyone.

Early majority: People from the general population who are the first to adopt new technology in the mainstream.

Late majority: People from another part of the general population who follow the early majority and start using the new things in their everyday lives.

Laggards: People who are slower to adopt new products and ideas. They are risk-averse and inflexible.

Each of the five types of adopters is affected by the five primary elements that influence innovation adoption, although to varying degrees: relative advantage, compatibility, complexity, trialability, and observability.

Relative advantage: The extent to which an innovation is perceived superior to the concept, product, or item it replaces.

Compatibility: The degree to which the innovation is consistent with the prospective adopters' values, experiences, and requirements.

Complexity: The difficulty in comprehending and/or implementing the innovation.

Trialability: The extent to which an innovation can be evaluated or experimented with prior to adoption.

Observability: The degree to which the innovation yields measurable outcomes.

Relevance in the pursuit of digital inequality: The theory may lack some ICT-specific modifications of components, but it is more robust to the diverse and constantly shifting complexities of technologies. Thus, it provides an understanding of the adoption of ICT over time or the diffusion of the Internet.

11.5.2 Unified Theory of Acceptance and Use of Technology (UTAUT)

In 2003, V. Venkatesh, M.G. Morris, G.B. Davis, and F.D. Davis proposed UTAUT by utilising popular models in the field of technology adoption. UTAUT is a popular framework for understanding how people adopt and use new technologies. It suggests that actual technology use is determined by behavioural intent. According to this theory, the perceived likelihood of adopting technology is directly influenced by four critical constructs: performance expectation, effort expectation, social influence, and enabling conditions. Age, gender, experience, and voluntariness moderate the effect of these indicators.

Components of the UTAUT include -

Performance expectation: This refers to the degree to which individuals perceive that a specific technology will assist them in performing their tasks more effectively or efficiently.

Effort expectation refers to the degree to which individuals assume that using a specific technology will be simple and require minimal effort.

Social influence: This refers to the extent to which individuals are influenced by the opinions and attitudes of others in their social networks, such as family, friends, and colleagues.

Enabling conditions: This refers to the extent to which individuals have access to the required resources and infrastructure, such as computers, internet connectivity, and technical support, to use a specific technology.

Relevance in the pursuit of digital inequality: Researchers have utilised the UTAUT to examine the factors that influence the adoption and use of digital technologies in communities with limited access in the context of the digital divide. Even though UTAUT has been widely utilised to characterise the digital gap, UTAUT and its components have been limited in characterising recent patterns of digital inequality.

11.5.3 Resources and Appropriation Theory

Jan van Dijk (2005) explained the resources and appropriation theory, addressing the social, cultural, and technological factors contributing to digital inequality.

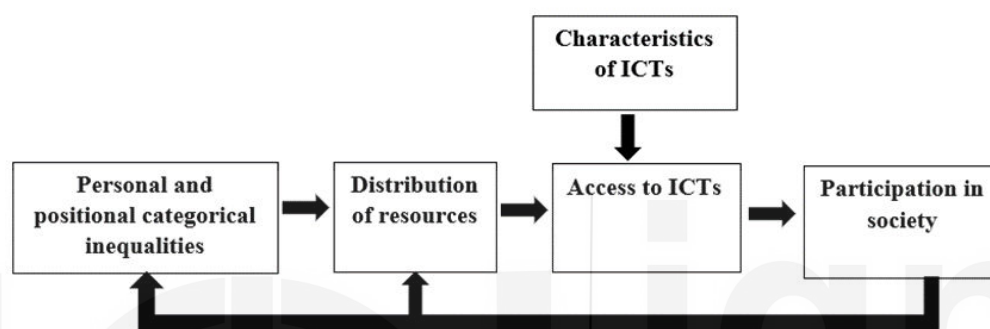


Figure 11.1 A Casual Model of Core Arguments

Source: Jan van Dijk (2005), *The Deepening Divide*

The core arguments of the theory are as follows:

- The unequal distribution of resources results from categorical inequalities in society.
- Unequal access to digital technologies results from an unequal distribution of resources.
- Unequal access to digital technologies is also influenced by the features associated with the technology.
- Unequal participation in society results from unequal access to digital technologies.
- Unequal distribution of resources and categorical inequalities are reinforced by unequal participation in society.

The components of the core arguments have been narrated as follows:

- Categorical inequalities:** Various personal and positional categories are accountable for the unequal distribution of resources required to access digital technologies. Personal categories comprise individual traits such as age, sex, race, intelligence (cognitive, emotional, and social), and personality (introvert, extrovert, and others). Positional categories include stratifications in the following areas: household (parent-child, husband-wife), management-executive, developing-developed, urban-rural, citizen-migrant, and high-low education.

- ii) **Resources and mechanisms of distribution:** An uneven access to digital technologies results from an unequal distribution of resources, where a few mechanisms (social exclusion, exploitation, and control) work throughout the process. The resources identified in theory are temporal (time to spend on different activities in life); material resources (income and all kinds of monetary properties); mental resources (knowledge, social, and technical skills, excluding digital skills); social resources (social network positions and relationships); and cultural resources (cultural status and all kinds of credentials).
- iii) **Successive kinds of access:** The fundamentals of resources and appropriation theory explain the stages of digital appropriation, with motivation access being the first, material access being the second, and usage access following after skill access.
 - **Motivational access:** Motivation refers to the factors, such as perceived utility and relevance, that influence people's interest in and engagement with digital technologies. Motivations vary from an attitude towards a particular digital medium to financial factors.
 - **Material or physical access:** Material or physical access refers to the physical accessibility of digital technologies such as computers and the Internet. Time (whether one has time or not) and places (home, school, public places, etc.) to use digital tools and technologies all come under the umbrella of physical access.
 - **Skills access:** Skill access refers to the digital literacy and skills required to utilise digital technologies and the Internet effectively. Skills that account for the usage of digital media encompass operational (the skills needed to operate computers and internet connections), informational (skills used to search, select, and process information from computer and network sources), and strategic (skills to achieve personal or professional goals).
 - **Usage access:** Usage access describes how individuals utilise digital technologies and the Internet, precisely, the intensity of usage time and diversity of applications.
- iv) **Properties of ICT (Hardware, Software, and Content):** The degree to which a person accesses a particular ICT depends on its technological characteristics. Some of the properties of an ICT tool facilitate access, while others limit it. Complexity and expense limit access, while multifunctionality and network effects expand it. The formats of new media content are an essential characteristic that has a more significant impact on accessibility.
- v) **Fields of participation in society:** In many parts of society, access to new media can mean the difference between being included and excluded. The umbrella term for these repercussions is participation in society. People with less exposure to digital media have steadily

decreasing opportunities to participate in arenas such as citizenship, education, politics, culture, social interactions, and health care.

Relevance in the pursuit of digital inequality: Researchers have used resources and appropriation theory to comprehend digital inequality by analysing the distribution of resources and how individuals and groups appropriate and utilise them in the digital domain.

11.6 CONCEPTUAL DIMENSIONS FROM A SOCIOLOGICAL PERSPECTIVE

Several traditional social theories serve as a background to the conceptual understanding of sociological approaches to digital inequality that can be used to probe its many facets and depths. Bourdieu's conceptions of capital and Weber's social stratification delve into the humanistic and social influences residing in digital inequality.

11.6.1 Bourdieusian Approach

Bourdieu (1986) defines capital as any resource that gives an individual an advantage that can be accumulated over time. Capital enables agents to replicate their positions within the social field and can take one of three fundamental forms: economic, cultural, or social capital, based on the arena in which it operates. In turn, each person's capital supports social inequality and social hierarchy. The conceptualisation of capital by Bourdieu has laid a foundation for a deeper understanding of digital inequality (Ragnedda & Ruijter, 2020).

In other words, people use and invest their social, cultural, and economic assets to strengthen their social status (symbolic capital). The capital of an individual not only creates the first level of the digital divide between those who can and cannot access the Internet but also the second level of the digital divide in terms of skill and capacity to use the Internet and the third level of the digital divide in terms of the social, economic, cultural, political, and personal advantages of being an internet user.

Bourdieu introduced the concept of social fields, structured spaces where individuals and groups compete for resources and power. In digital inequality, the digital realm can be considered a social field where individuals' engagement is influenced by their habitus—the internalised set of dispositions, behaviours, and preferences shaped by their social and cultural background. Digital inequality through the lens of social fields and habits allows an understanding of how individuals' social positioning and cultural capital influence their access to and use of digital technologies.

11.6.2 Weberian Approach

The Weberian approach, rooted in the sociological theories of Max Weber, can provide valuable insights into the intertwined processes between social and digital inequalities. Social inequalities as a potential topic of discussion

in the discipline of digital inequality have been examined through the lens of Weber's social stratification and power dynamics. The Weberian approach to digital inequality uncovers the structural and systemic factors that contribute to digital inequality and guides efforts to mitigate disparities in access, skills, and resources in the digital realm.

Weber emphasised social stratification based on various dimensions, such as class, status, and power. Social stratification illuminates the manifested and circulated uneven distribution of resources. As per the concept, a group of people, being in a privileged position in society, take more advantage of ICTs and put them to use in achieving a tangible outcome (enhancement of life chances), such as job search, purchase, access to health care, political engagement, learning, socialisation, cybersecurity, leisure, and relationships with bureaucracy. As a result, different social groups, such as economically disadvantaged or marginalised communities, experience different levels (high to low) of access and benefits in the digital realm.

According to this approach, digital stratification intensifies on the following grounds:

Class: A class with limited capacity to use ICTs lacks the economic opportunities they offer.

Status group: A status group with limited capacities to use ICTs cannot serve the benefits of usage in status building, enhancement, and improvement.

Power: Limited capacities to use ICTs delimit personal interests, political power, and influence.

11.7 DIGITAL CAPITAL

Digital capital is the Bourdieusian approach to the digital realm. Following Bourdieu's concept of capital, digital capital is defined as the accumulation of both digital competence (internal skills and attitudes) and digital devices (external resources). It improves the tangible outcome of digital technologies to turn them into other forms of capital, such as social, economic, and cultural.

The core arguments for the concept are as follows:

- Digital capital is closely connected with economic, social, and cultural capital. A person needs these three capitals to build a good attitude and the right skills to use technology well.
- Digital capital is distributed unequally in society. Individuals with higher traditional forms of capital (such as education and social prestige) tend to have higher levels of digital capital. This indicates that people from more privileged backgrounds are more likely to have the skills, resources, and social connections necessary to flourish in the

digital world. In contrast, less privileged people may struggle to access and utilise digital resources effectively.

- People need a network of support (social capital) like friends, family, and an online community to build up their digital capital. At the same time, people need money (economic capital) to access ICTs and knowledge, skills, and abilities (cultural capital) to use ICTs effectively.

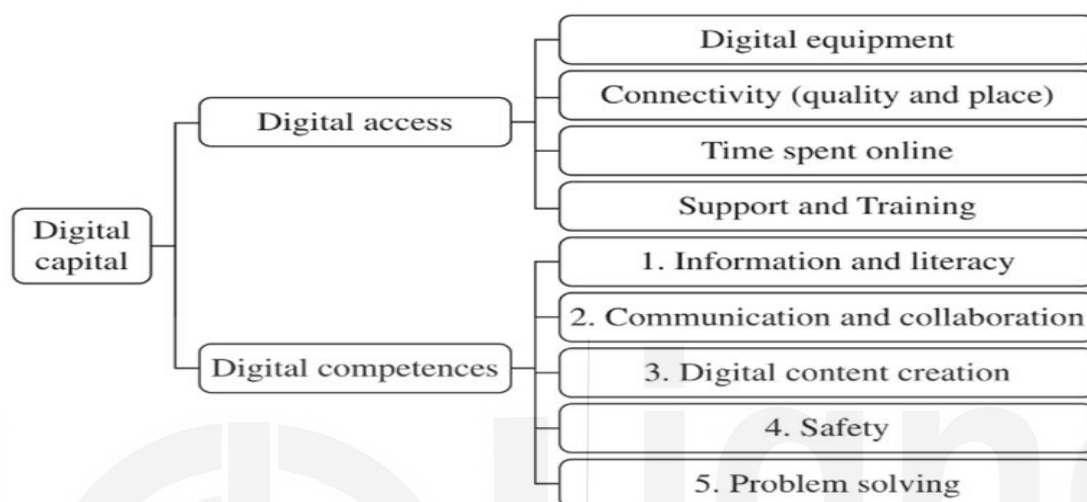


Figure 11.2 The Components of the Digital Capital Model

Source: Massimo Ragnedda and Maria Laura Ruiu (2020), Digital Capital: A Bourdieusian Perspective on the Digital Divide.

In 2020, Massimo Ragnedda and Maria Laura Ruiu proposed the model of digital capital. Digital competence and digital access are two distinct components of digital capital that can be built up and transferred to improve life chances. According to the concept, in an Internet-based society, digital capital joins the Bourdieusian capitals as an essential factor in illuminating the causes and consequences of social stratification and differentiation.

An individual's digital capital can be accessed by cumulating all the elements of digital access and digital competence. This model helps comprehend the patterns of digital inequality concerning the resourcefulness of digital capital. In simple words, it identifies digital capital inequality.

The elements of the core components have been described below:

Digital Access

Four elements of digital access:

- i) **Digital equipment:** Devices used to access the Internet, such as mobile phones or smartphones, laptops, and desktop computers.
- ii) **Connectivity:** Access quality depends on which setting is used to access the Internet.
- iii) **Time spent online:** Cumulative experience in using ICTs in terms of total time of internet usage (years of usage)

- iv) **Support:** Seeking help to support experiences related to digital technology

Digital competence

i) **Competence Area 1: Information and Literacy**

- The activities of browsing, searching, filtering, data, information, and digital content
- Evaluating data, information, and digital content
- Managing data, information, and digital content

ii) **Competence Area 2: Communication and Collaboration**

- Interacting through digital technologies
- Engaging in citizenship
- Sharing and collaborating through digital technologies
- Managing digital identity

iii) **Competence Area 3: Digital Content Creation**

- Developing, integrating, and re-elaborating digital content
- Knowledge about copyright and licences
- Programming

iv) **Competence Area 4: Safety**

- Protecting devices, personal data and privacy
- Protecting health and well-being and the environment

v) **Competence area 5: Problem-solving**

- Solving technical problems, identifying needs and technological responses
- Identifying digital competence gaps

11.8 DIGITAL INCLUSION - EXCLUSION AND PARTICIPATION

Digital inclusion refers to the efforts and initiatives aimed at ensuring that all individuals, regardless of socioeconomic background, geographic location, age, gender, or ability, have equal access to and opportunity to use digital technologies effectively. Digital inclusion efforts may involve:

- Increasing internet access
- Bridging the digital skills gap
- Access to digital devices

- Creating digital content
- Addressing affordability
- Empowering marginalised communities
- Policy and regulatory interventions

Digital exclusion is when individuals or groups are systematically excluded from accessing and utilising digital technologies and resources, resulting in a lack of participation in the digital world.

The digital participation of individuals assures the digital inclusion of an individual and, on a larger scale, a specific group or community. In this context, participatory inequality, which may be defined as people's inability to use the Internet to produce and disseminate digital resources, is a potential barrier to the growth of a network society.

Table 3. Areas of participatory inequality

Area	Inequality
Economic participation	Inequalities regarding the usage of ICTs to participate in the labour market and business
Educational participation	Inequalities regarding educational usage of ICTs
Social participation	Inequality in the field of social capital building (e.g., building social networks)
Cultural participation	Inequality related to online cultural practices
Political participation	Inequality over the Online political discourse
Institutional participation	Inequality in achieving digital citizenship

Source: Adapted from Jan van Dijk (2005)

To eradicate digital exclusion, stakeholders (governments, organisations, and local groups) must work together. Efforts to close the digital gap include building internet infrastructure, making devices and connections more affordable, providing digital skills training and literacy programmes, and creating digital platforms open to all users and meeting their different needs. These actions aim to ensure that everyone has the same chance to access and benefit from the digital world.

11.9 NEW FORMS OF DIGITAL INEQUALITY

The rapid advancement of technology has resulted in the emergence of new types of digital inequality, discussed as follows:

Privacy and digital inequality: Privacy concerns are more important than ever in the digital age. Those with more information, resources, and technological skills can protect their personal data by implementing privacy-enhancing methods and practices. On the other hand, people with little access

to resources or digital literacy may be more at risk of privacy violations, data exploitation, or data hacking, which can increase inequality.

Algorithms and digital inequality: Everything from search engine results to automated decision-making systems depends heavily on algorithms. Internet users consciously or unconsciously deal with unseen infrastructure called an algorithm, logical rules, and technical prescriptions developed by programmers, mathematicians, and data engineers. Algorithms are essential for creating favourable conditions for participation in digital life; therefore, digital inequality arises from a lack of understanding of algorithms. Lack of awareness could negatively affect democracy and public involvement on a societal level.

Artificial intelligence (AI) and digital inequality: Automation and other emerging technologies like artificial intelligence (AI) have the potential to produce new types of inequality. Access to and proficiency with these technologies can lead to increased productivity, better career possibilities, and financial gains. However, those people and communities that do not have the digital literacy to adapt to these disruptive technologies may be excluded from these advantages, which would increase inequality.

Check Your Progress: 2

Note: 1) Use the space below for your answers.

2) Compare your answers with those given at the end of this unit.

1. Explain the Weberian approach to digital inequality.

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2. What are the core arguments of “Resources and Appropriation Theory”?

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3. Write the definition of digital inclusion.

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4. What are the two components of the digital capital model?

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11.10 LET US SUM UP

In this Unit, we gained a comprehensive understanding of digital inequality. It explained the factors contributing to inequality, such as disparities in access to technology, internet connectivity, digital skills, and usage factors. Delving into diverse frameworks of digital inequality, it defined the patterns and manifestations of digital inequality across different dimensions of academic disciplines. It also sheds light on how digital disparities intersect with factors like age, gender, race, ethnicity, income, and geographical location, how unequal access to digital resources and skills affects opportunities and outcomes, and possibilities to ensure equitable and secure digital environments as well. We learnt that the challenges faced by digital exclusion contribute to broader social and economic inequalities. We also explored new inequalities such as privacy, algorithmic, and AI-related inequalities. The discussion provided a deep understanding of the concept, offered diverse perspectives, and equipped you with knowledge and insights from a technological and sociological perspective of digital inequality.

11.11 KEY WORDS

Internet inequality : Internet inequality refers to the differences between people, communities, and nations regarding their capacity to access and efficiently use the Internet.

Digital inclusion : Digital inclusion refers to the inclusion of individuals and communities in the digital world, which means that they have access to digital technologies, internet connectivity, and the necessary skills to utilise them effectively and thus access to the economic, social, and cultural benefits of the digital world.

ICT : Information and communication technology (ICT) comprises diverse hardware, software, networks, and digital systems that allow for the processing, storage, transmission, and retrieval of data and information.

Digital participation : Digital participation is the active engagement, involvement, and inclusion of individuals and communities in the digital world through digital technologies to access information, communicate, create, collaborate, and contribute to digital content and platforms.

AI : Artificial intelligence (AI) refers to emulating human intelligence in machines programmed to carry out tasks that typically require human intelligence. It allows machines to perceive, reason, learn, and solve problems, enabling them to imitate or replicate human cognitive abilities.

11.12 FURTHER READINGS

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11.13 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress: 1

1. The term "digital inequality" reflects a broader conception of the digital divide and incorporates a variety of factors beyond access to technology, including skills, usage patterns, and stakeholders involved. It acknowledges that merely providing access (binary access to digital technologies) to technology is insufficient to ensure equal benefits from the opportunities it provides.
2. Three levels of the digital divide correspond to multiple dimensions of digital inequality. The first level of the digital divide describes the unequal distribution of access to digital technologies. The second level of the digital divide refers to the unequal distribution of digital skills and competence among people with access to digital technologies. The third level of divide refers to the disparities in individuals' abilities to use digital technologies to achieve tangible outcomes of technology usage.

3. Determinants of digital inequality include economic determinants - income, employment status, employment type, occupational status, and educational level; social determinants - online social interaction, social networking, types of social activities; cultural determinants - religion, ethnicity, and internet use language.
4. Inequality in the information society appears at the most fundamental level of information need, which leads to a widening gap in the capacity for association and knowledge transfer and the capacity to judge the quality of information and make other information-related decisions. In modern times, there is a considerable disparity in the distribution of information abilities among different groups of people.

Check Your Progress: 2

1. Social inequalities as a potential topic of discussion in the discipline of digital inequality have been examined through the lens of Weber's social stratification and power dynamics. The Weberian approach to digital inequality uncovers the structural and systemic factors that contribute to digital inequality and guides efforts to mitigate disparities in access, skills, and resources in the digital realm.
2. Core arguments of the Resources and Appropriation Theory are (i) Unequal distribution of resources results from categorical inequalities in society. (ii) Unequal access to digital technologies results from an unequal distribution of resources. (iii) Their features also influence unequal access to digital technologies. (iv) Unequal social participation results from unequal access to digital technologies.
3. Digital inclusion refers to the efforts and initiatives to ensure that all individuals, regardless of socioeconomic background, geographic location, age, gender, or ability, have equal access to and opportunity to use digital technologies effectively.
4. Digital competence and digital access are two distinct components of digital capital that can be built up and transferred to improve life chances.

UNIT 12 DIGITAL MEDIA AND ENVIRONMENT SUSTAINABILITY

Structure

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- 12.1 Learning Outcomes
- 12.2 Understanding ICT and Its Lifecycle Impact on Environment
 - 12.2.1 ICT in the Production Phase
 - 12.2.2 ICT in the Use Phase
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 - 12.4.2 Smart Cities and Sustainable Transportation
 - 12.4.3 Energy Efficiency and Green Computing
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12.0 INTRODUCTION

Have you ever wondered how your online activities impact the environment? Or when you discard an electronic device, you are contributing to the surging e-waste. Even by searching a single query in the Google search engine, you are releasing 0.2 grams of CO₂ into the atmosphere. (Hölzle, 2009)

However, it is worth mentioning that you alone are not to be blamed. Maybe the technology and the device were designed in such a way that it would impact the climate. Planned obsolescence and the disposable nature of older technologies in the advent of new technologies are making the impact of the ICT sector on climate very much visible.

Though this impact is minuscule compared to other sectors' carbon emissions like manufacturing, energy, transportation and aviation; the increasing penetration of digital devices and heavy dependence on IT cannot be ignored.

The ICT sector's carbon footprint currently poses at nearly 2 per cent of overall global. It could be further reduced by over 80 per cent if all electricity consumed came from renewable energy sources or switching devices less frequently.

If used well, digital technologies can also help in decarbonization in other sectors. Smart use of ICT can reduce the impact on other sectors by up to 16 per cent or more. (Hankel, 2014)

Integration of ICT to create solutions that enhance resource efficiency and reduce environmental impact can contribute to a more sustainable, resilient future and environment sustainability.

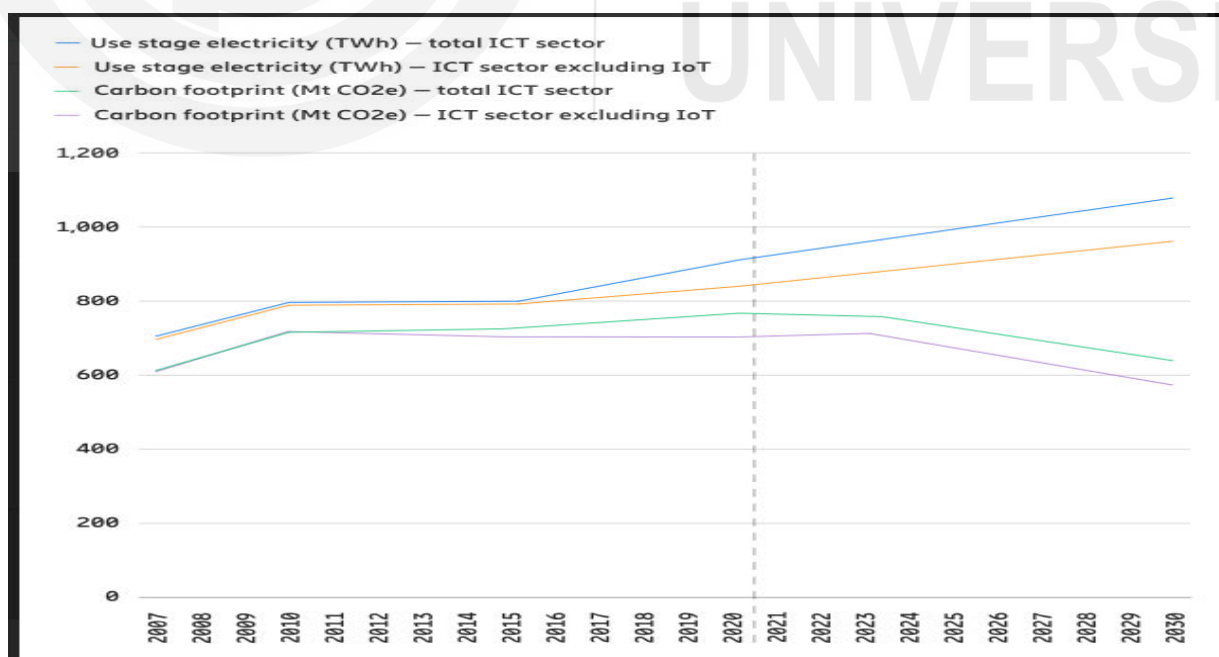


Figure 12.1: ICT sector development 2007–2020 and forecast to 2030

Source: ericsson.com/mobility-report.

12.1 LEARNING OUTCOMES

After working through this Unit, you will be able to:

- understand the meaning of ICT and sustainable development;
- know about Green ICT, its pros and cons, transformations, and challenges; and the need for its integration;
- identify the role of different stakeholders in adopting sustainable development; and
- learn about the future roadmap of development through Green ICT and our role in it.

12.2 UNDERSTANDING ICT AND ITS LIFECYCLE IMPACT ON ENVIRONMENT

Information and communication technology (ICT) serves as a broad term encompassing communication devices and applications such as radio, television, cellular phones, computer hardware, software, satellite systems, and associated services like videoconferencing and distance learning. Different studies employ varied terms like "ICT," "ICT products," "ICT solutions," "ICT goods," and "ICT services."

European statistics (Eurostat 2016) defines "ICT" as all technical means for handling information and supporting communication. The OECD (2011) uses the term "ICT products and services," covering an extensive range of categories. ICT can be categorised broadly in two ways: as ICT solutions (in terms of products and services) and as ICT societies, where ICT plays a vital role in societal life and development, considering all ICT solutions.

Many scientific studies propose that ICT can facilitate the transition to a less material-intensive economy, promoting sustainability (Hilty et al. 2011). The Smarter2020 report (GeSi 2012) outlines ICT's potential role in reducing future energy use and climate change impact through digitalization, dematerialization, data collection, communication, systems integration, demobilization, mass customization, intelligent operation, and soft transformation and optimization processes across various sectors.

However, ICT usage entails various implications also. The environmental impacts of ICT can be categorized into different orders. As per Berkhout and Hertin (2001), there are direct, indirect, and structural/behavioural effects.

Direct effects: These are negative effects; that arise from the manufacturing, use, and disposal of ICT devices. Eg. E-waste, and resource depletion.

Indirect effects: These effects are both negative and positive. They result from ICT applications, such as dematerialization and efficiency gains. E.g. travel substitution transportation optimization and cultural homogenization.

Behavioural/ Structural effects: These are related to lifestyle and structural changes caused by ICT, encompassing both positive and negative aspects, including the rebound effect. It suggests that as ICT improves efficiency, reducing resource consumption or environmental impact per unit of service (e.g., energy-efficient devices), the overall demand for that service may increase. In the context of ICT, it means that increased efficiency might lead to more extensive use and, consequently, a potential increase in resource consumption. This is also known as Jevons' Paradox, named after the economist William Stanley Jevons.

Some other examples are changed communication patterns, information overload, digital addiction, the digital divide loss of jobs and the shift to the service economy. It reshapes how we lead our lives on a societal level.

Apart from these factors, the utilization of information and communication technology (ICT) contributes to its carbon footprint. The carbon footprint concept extends beyond the electricity consumption of ICT products, encompassing the greenhouse gas emissions linked to the energy and materials utilized across the entire product life cycle. This encompasses activities such as acquiring raw materials, production and assembly, transportation, operational use, and end-of-life treatment. Notably, the indirect consequences of ICT usage are not accounted for in the carbon footprint calculation, despite frequently being more substantial than the footprint itself.

The life cycle of Information and Communication Technology (ICT) products (PCs, computer screens, mobile phone networks, fixed-line networks, components like ICs and batteries) and "green data centres" (that store, manage and distribute data) consists of three phases: **production, use, and end of life**, with the **design phase** influencing the life cycle substantially.

Each phase has its life cycle. Each phase has environmental impacts in terms of consumption of materials and energy. The major energy consumption is, primarily in the use phase. And, this perspective is criticized for oversimplification due to the following reasons:

- The mix of electricity sources matters for energy generation.
- Phases beyond usage might be more crucial.
- Other environmental effects may be more significant than just energy consumption.

12.2.1 ICT in the Production Phase

Early studies on the ecological impacts of computers highlight the significance of the production phase in comparison to the usage phase. A recent study on desktop PC production in China [Eugster et al., 2007] found

that the energy demand for producing a PC equals about three years of typical usage.

It takes roughly six years of use for the usage phase to balance out the production phase. This comparison becomes even more favourable for the usage phase if the energy supply includes a significant share of non-fossil sources (e.g., hydro, wind, or solar power). The production of integrated circuits and parts with precious metals is identified as the most energy-intensive aspect of PC production. Assembling the parts has a comparatively minor environmental impact.

The production phase of Information and Communication Technology (ICT) has a substantial environmental impact, involving resource consumption, energy use, and waste generation. Let us glance over some examples illustrating the environmental impact of ICT production:

High Energy Consumption in Manufacturing Processes: The production of integrated circuits (ICs) and semiconductor chips, essential components in electronic devices, involves complex manufacturing processes requiring high energy consumption and the use of various chemicals.

Mining for Rare Earth Metals: The extraction of rare earth metals, such as neodymium, used in the production of magnets for hard drives and other electronic components, contributes to habitat destruction and environmental pollution. Rare earth mining is associated with soil and water contamination.

E-waste Generation: The production phase contributes to electronic waste (e-waste) generation. Improper disposal and recycling of electronic components lead to environmental pollution. For instance, improper disposal of printed circuit boards can release harmful substances like lead and brominated flame retardants into the environment.

Carbon Emissions: The energy-intensive processes involved in manufacturing ICT components, including the production of devices like computers and smartphones, contribute to carbon emissions. The use of fossil fuels in manufacturing plants increases the carbon footprint of ICT products.

Packaging Materials Impact: The packaging materials used in transporting and protecting ICT products contribute to the environmental impact. Excessive use of non-biodegradable packaging materials adds to the overall ecological footprint.

Global Supply Chain Impact: The global nature of ICT supply chains involves the transportation of components across long distances, contributing to carbon emissions. The transportation of raw materials and finished products adds to the environmental impact.

Water Usage: Water is a crucial resource in the production phase, particularly in semiconductor manufacturing. The use of water-intensive

processes, such as cleaning and cooling, can lead to water scarcity issues in certain regions.

Chemical Usage: The production of electronic devices involves the use of various chemicals, including solvents and cleaning agents. Inappropriate disposal or mishandling of these chemicals can lead to soil and water contamination.

It is important to address these environmental impacts through sustainable practices, such as eco-friendly manufacturing processes, responsible sourcing of raw materials, and effective e-waste management and recycling initiatives.

12.2.2 ICT in the Use Phase

In the use phase, an ICT product provides its service to the user. Life Cycle Assessment (LCA) studies measure environmental impact based on a functional unit, like "1 hour of PC use." The key environmentally relevant factor during ICT use is the device's useful life, considering its substantial production impact. Using a device for a more extended period, say six years instead of three, reduces the production impact per functional unit by half.

Generally, the longevity of ICT devices matters more than their power consumption during use. However, in cases like servers running 24/7, energy consumption during use becomes crucial. Short software innovation cycles can harm hardware life, negatively impacting the environment. Extracting precious metals, especially gold, for electronics significantly contributes to their overall environmental impact. The use phase of Information and Communication Technology (ICT) products involves their operation and energy consumption during normal usage. Here are some examples illustrating the environmental impact of ICT in the use phase:

Energy Consumption: The operation of ICT devices, such as computers, smartphones, and servers, requires a significant amount of energy. For instance, data centres that host online services consume vast amounts of electricity.

Carbon Emissions: The electricity used to power ICT devices often comes from fossil fuel-based sources. This contributes to carbon emissions and exacerbates climate change. The carbon footprint depends on the energy mix of the region where the devices are used.

Electronic Waste (E-waste): The use phase contributes to electronic waste generation, especially when devices become obsolete or are discarded. Devices with short lifespans or those that are not easily upgradable contribute to a faster turnover of electronic products.

Phantom Energy Consumption: Many ICT devices consume energy even when not in active use, known as phantom or standby power. For instance, leaving a computer or television on standby mode can lead to unnecessary energy consumption over time.

Internet Infrastructure Impact: The use of online services, streaming platforms, and cloud computing contributes to the overall energy consumption of the Internet infrastructure. Data transmission, storage, and processing in data centres have environmental implications.

Short Device Lifespan: The relatively short lifespan of some ICT devices, driven by rapid technological advancements and changing consumer preferences, can result in more frequent replacements. This leads to increased production-related environmental impacts and e-waste.

Cooling Systems: High-performance ICT equipment, especially in data centres, requires extensive cooling systems to prevent overheating. These cooling systems consume additional energy, contributing to the overall environmental impact.

Resource Depletion: The use of certain minerals and metals in ICT devices, such as coltan in smartphones, can contribute to resource depletion. Unsustainable mining practices for these materials can lead to environmental degradation.

Emission from Network Infrastructure: Example: Telecommunication networks, including mobile networks and broadband, contribute to energy consumption and emissions. The energy usage of base stations and the transmission of data contribute to the overall impact.

Addressing the environmental impact in the use phase involves promoting energy efficiency, designing products for longevity and upgradability, and encouraging responsible consumer behaviour, such as proper disposal and recycling of electronic devices.



Fig. 12.2

Delhi-based Namo E-waste Management, a recycler cum PRO(Producer Responsibility Organisation), extracts metals such as aluminium, copper, brass and zinc from e-waste products that include computers, motherboards, mobile phones and server boards (source Indian Express)

12.2.3 ICT in the Disposal Phase

The disposal phase of an ICT product, often referred to as the end-of-life phase, involves the proper discarding or recycling of the product. During this

phase, the environmental impact is influenced by the methods used for disposal. Examples of ICT environment impact in the disposal phase include:

Improper E-Waste Disposal: Discarding electronic devices in landfills without proper recycling leads to leaching of hazardous materials into soil and water, posing environmental and health risks.

Informal E-Waste Recycling: Unregulated recycling practices, common in some developing countries lead to health hazards for workers, and environmental pollution due to inadequate processing methods.

The recycling of electronic waste is a significant and growing concern globally due to the increasing volume of Waste Electrical and Electronic Equipment (WEEE). Despite methodological challenges in assessing environmental impacts, recycling e-waste is shown to be environmentally beneficial, saving around 20-25% of energy consumed in the production phase by recovering metals. However, the environmental benefits are contingent on proper recycling processes, and informal recycling, prevalent in some emerging economies, poses health and environmental risks. Many countries are implementing WEEE recycling systems, but challenges arise with the integration of ICT products into everyday objects, creating new recycling challenges. There is a huge informal industry in India which recycles WEEE mainly to recover gold and copper. It is deemed profitable if the health and environmental impacts are not accounted for.



Figure 12.3 shows the extraction of copper from printed wiring boards in a typical backyard company with 12 workers in Delhi. This company recovers 1-2 tons of copper per month [Widmer et al., 2005].

12.3 KEY PILLARS OF SUSTAINABLE DEVELOPMENT

As per Ismail Serageldin (2006), “Sustainability is giving future generations as many opportunities as, if not more than, we have had ourselves.”

Sustainable development is a holistic approach that integrates economic, social, and environmental considerations to create a balanced and enduring future. It refers to a mode of growth or progress that meets the needs of the present without compromising the ability of future generations to meet their own needs. It encompasses:

12.3.1 Ecology

Environmental Protection is the most important pillar. It acknowledges a responsibility to care for and protect the environment by implementing practices that ensure the sustainable use of natural resources.

It aims to protect and preserve the natural environment, biodiversity, and ecosystems for current and future generations. It involves sustainable resource management, conservation efforts, pollution reduction, and measures to combat climate change.

12.3.2 Economics

We cannot thrive without Economic Sustainability. It focuses on promoting economic growth that is inclusive, equitable, and contributes to the overall well-being of society. Emphasizes responsible and sustainable business practices, fair trade, and the efficient use of resources to support long-term economic viability and foster innovation and adopting practices that enhance economic resilience.

However, creating eco-friendly products or services faces a challenge because they often end up supporting the same market-driven system they aim to improve. This is called "The Paradox of Green Consumerism".

The world emphasizes sustaining economic growth, but many argue that continuous growth is not compatible with environmental well-being. When green initiatives operate within existing market structures, there's concern that they may reach a point where they can't effectively lead us towards a truly sustainable future.

12.3.3 Ethics

Ethics are crucial in the context of sustainability. They provide guiding principles and moral standards that help individuals and organizations make decisions aligned with sustainable practices. They set a moral compass and ensure that actions taken for sustainability are not merely short-term solutions but contribute to the long-term well-being of individuals, communities, and the environment.

Ethics recognizes the impact of human activities on the environment and takes responsibility for minimizing negative effects. They hold individuals, organizations, and governments accountable for their environmental actions.

Ethical consideration encourages individuals to make ethical choices in their consumption patterns and supports environmentally friendly products and practices. It concentrates on ensuring social equity, justice, and ethical well-being for all members of society. It addresses issues such as poverty, education, healthcare, human rights, and social inclusion to create a just and cohesive community.

It fosters inter-generational responsibility; considering the needs of present and future generations in decision-making. Strive for a balanced and harmonious coexistence with the environment. Enhance Global Cooperation: Recognizing the interconnectedness of global challenges.

It encourages collaboration and cooperation among nations to address shared sustainability goals. Encourages transparency by providing clear and accurate information about environmental practices. Allows stakeholders to make informed decisions and hold entities accountable. Promotes community engagement by involving communities in decision-making processes related to environmental sustainability. Respecting and considering local knowledge and values.

Check Your Progress: 1

Note: 1) Use the space provided below for your answers

2) Compare your answers with those given at the end of this unit.

1. Design a Green ICT initiative for a local community or organization. Consider aspects such as energy efficiency, waste reduction, and sustainable practices. Additionally, discuss how this Green ICT initiative aligns with broader environmental sustainability goals.

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12.4 GREEN ICT AND ITS NEED FOR ENVIRONMENT SUSTAINABILITY

Historically, technology, including IT, has acted as a "force multiplier," allowing humans to achieve greater impact with less effort, similar to the invention of agricultural implements or powerful weapons. ICT effectively compresses time by storing vast information, modelling the past, predicting the future, and synchronizing human activities. It also compresses space through long-distance communication, global mapping, and facilitating worldwide transportation of goods and people. Additionally, ICT simplifies complexity by enhancing memory, performing calculations, and establishing cooperation standards.

In the context of human institutions, ICT serves as a bridge between different scales, providing individuals with accessible information about institution-scale phenomena. It enables institutions to make informed decisions based on expert opinions, computational simulations, modelling, and visualization. By connecting human, institutional, and environmental scales, IT empowers individuals to impact environmental concerns through institutional mediation.

The intersection of environmental issues and ICT is known as Green ICT. The term "Green" here signifies an awareness of the environmental impact of

human activities, symbolized by the colour of plant leaves. It helps to make all aspects of society greener, including IT itself.

It explores the potential for ICT to address global ecosystem issues. While technology has historically contributed to environmental issues, Green ICT focuses on mitigating these problems and promoting sustainable living.

It thrives on the convergence of two prominent trends: the growing global concern for environmental issues and the swift expansion of information technology (ICT).

The first trend involves an increasing awareness of environmental issues. The second trend focuses on the widespread proliferation of ICT globally as once a rarity, mobile phones are now owned by over half of the world's population.

The primary goal of Green IT is to spread information about environmental concerns and enable individuals and organizations to adopt best practices. Existing examples of environmental ICT systems include smart energy grids and systems optimizing hybrid car engines, GPS and online mapping software. It promotes efficient travel and reduces CO₂ emissions.

Green ICT can be categorized along the axis of "personal" and "institutional" applications.

- Personal IT systems empower individuals and small groups to enhance their understanding and response to environmental concerns.
- Institutional IT systems extend this impact to larger organizations, including corporations, governments, NGOs, and universities.

These two forms of Green IT work together to address environmental challenges and promote sustainability.

However, we should not forget the fact that Green ICT is not a panacea, it may be able to help but it is not a magic wand. "Technology will not save the world; people will. New forms of IT, different from the growth technologies will help to solve the current untenable issues." Tomlinson, 2010.

12.4.1 ICT Innovations and Sustainable Solutions

The potential opportunities in ICT and Sustainable Solutions are significant. Information and Communication Technology (ICT) can extend healthcare services to remote rural areas, enhance the efficiency of electricity grids, and optimize traffic flow. ICT solutions, incorporating technologies like IoT, machine learning, and automation, hold substantial promise for globally reducing carbon emissions across various sectors and industries. Current ICT solutions are estimated to have the potential to decrease global carbon emissions by up to 15 per cent, equivalent to about one-third of the crucial emission reduction target by 2030.

The advent of new technologies such as 5G, IoT, and AI will create additional possibilities for emission reduction. The realization of these opportunities and the avoidance of adverse use will play a crucial role in the world's efforts to limit global warming.

The integration of Information and Communication Technology (ICT) in various domains contributes to sustainable solutions and environmental conservation. Key applications include the implementation of IoT (Internet of Things) devices for smart city technologies, optimization of renewable energy integration, digital platforms for circular economy practices, blockchain for supply chain transparency, AI-driven environmental monitoring, eco-friendly computing, digital twins for sustainability, precision agriculture with drones and sensors, ICT in conservation using remote sensing and satellite imaging, and the incorporation of ICT in green transportation solutions for traffic optimization and promoting sustainable modes of transportation.

Check Your Progress: 2

Note: 1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of this unit

1. What role can ICT play in promoting inclusive and equitable access to education, healthcare, and other essential services, particularly in developing regions, to support sustainable development?

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12.4.2 Smart Cities and Sustainable Transportation

The concept of "Smart Cities" has emerged globally due to rapid urban growth, rural-urban migration, and socio-economic divides. The term "smart" is synonymous with connected, resilient, intelligent, or digital communities.

Key components of smart cities include interoperability, sustainability, city-wide connectivity, security, effective transportation, and public-private partnerships.

India is rapidly adopting the concept as its urban population is expected to contribute significantly to the country's GDP. Smart cities incorporate competition, capital, and sustainability, providing good infrastructure and open access to amenities.

The environment plays a critical role in urban safety and greenery. Urbanization and climate change are converging, posing threats to urban regions globally. Despite challenges, cities have the potential to be centres of attraction for social and economic development.

Resilience in urban societies depends on factors like climate, land use, human values, and policies. Climate-resilient smart cities require disaster-resistant infrastructure and response capacities.

The focus of smart cities is to become citizen-centric and environmentally friendly with the help of technology. With the help of smart city technologies using IoT (Internet of Things) devices and sensors efficiency can be increased in planning, implementation, energy consumption, waste management, transportation, and overall urban planning.

Transportation systems with ICT integration can help in optimizing traffic flow, reduce emissions, and promote sustainable modes of transportation, such as electric vehicles and shared mobility services.

12.4.3 Energy Efficiency and Green Computing

Energy efficiency and green computing are essential aspects of sustainable IT practices. An example of energy efficiency in computing is the implementation of virtualization technologies. Virtualization allows a single physical server to run multiple virtual machines, each operating as an independent server. This helps in the optimal utilization of server resources, reducing the need for additional physical servers and, consequently, lowering energy consumption.

Green computing also involves using energy-efficient hardware components. For instance, companies and data centres may opt for energy-efficient processors, solid-state drives (SSDs) that consume less power compared to traditional hard disk drives (HDDs), and energy-efficient power supplies.

Moreover, cloud computing services contribute to energy efficiency by consolidating data centres and enabling resource-sharing among multiple users. Cloud providers optimize server utilization, leading to lower energy consumption per user.

Additionally, power management features in computers and data centres, such as dynamic voltage and frequency scaling (DVFS), help adjust the power usage based on the workload. This ensures that systems consume only the necessary amount of power at any given time.

These examples showcase how energy efficiency and green computing practices can significantly reduce the environmental impact of IT operations.

12.4.4 Smart Grids and Energy Management

Smart grids and energy management systems play a crucial role in optimizing energy distribution and consumption. Here are examples of their applications:

Advanced Metering Infrastructure (AMI): Smart meters are deployed to collect real-time data on energy consumption. Consumers can monitor and manage their energy usage more efficiently, and utilities can gain insights into load forecasting and grid optimization.

Demand Response Programs: Automated systems that adjust energy usage in response to demand fluctuations or price signals. It reduces peak demand, enhances grid stability, and allows consumers to save on energy costs by adjusting consumption during peak times.

Grid Automation: Utilizes sensors, communication networks, and control systems to automate grid operations. It improves grid reliability, reduces outage durations, and enhances the integration of renewable energy sources.

Distributed Energy Resources (DERs) Integration: Integrates renewable energy sources, energy storage, and electric vehicles into the grid. It maximizes the use of clean energy, improves grid resilience, and allows for better management of distributed energy generation.

Energy Storage Systems: Batteries and other storage technologies used to store excess energy during low demand for use during high demand periods. It balances supply and demand, supports intermittent renewable sources, and enhances grid stability.

Home Energy Management Systems (HEMS): Smart devices and systems that allow homeowners to monitor and control their energy usage. Enables consumers to make informed decisions, optimize energy consumption, and save on electricity bills.

Grid Analytics:

Electric Vehicle (EV) Charging Management: Smart charging infrastructure that optimizes the charging process for electric vehicles.

12.4.5 Remote Collaboration and Telecommuting

Remote collaboration and telecommuting have become integral components of modern work culture, especially with the advancement of Information and Communication Technology (ICT). Here are some examples illustrating the application of remote collaboration and telecommuting:

Video Conferencing Platforms: Reduces the need for physical presence, enhances communication, and enables real-time collaboration. **Eg.** Platforms such as Zoom, Microsoft Teams, and Cisco Webex facilitate virtual face-to-face meetings, fostering collaboration among remote teams.

Cloud-Based Project Management Tools: Facilitates remote coordination, ensures transparent workflow management, and keeps team members on the same page. **Eg.** Tools like Asana, Trello, and Jira provide centralized platforms for project management, allowing teams to collaborate on tasks and projects.

Document Collaboration Platforms: Enhances document sharing and editing, supports version control, and ensures seamless collaboration on shared files. **Eg.** Google Workspace, Microsoft 365, and Dropbox enable real-time collaboration on documents, spreadsheets, and presentations.

Virtual Whiteboards: Facilitates visual collaboration, allows for remote ideation sessions, and ensures team engagement in creative processes. Eg. Tools like Miro and Microsoft Whiteboard offer digital platforms for brainstorming and collaborative drawing.

Instant Messaging and Team Chat Apps: Supports quick communication, reduces email overload, and enhances team connectivity regardless of physical location. Eg. Apps like Slack, Microsoft Teams, and Telegram provide real-time messaging and collaboration channels for teams.

Telecommuting Software Suites: Assists in monitoring productivity, ensures accountability in remote work, and provides insights into team performance. Eg. Software suites like RemoteHQ and Hubstaff offer various tools for time tracking, virtual collaboration, and remote team management.

Augmented Reality (AR) for Remote Assistance: Facilitates remote troubleshooting, improves training processes, and supports hands-on tasks without physical presence. Eg. AR tools like Microsoft HoloLens enable remote experts to assist by overlaying digital information onto physical objects.

Virtual Reality (VR) for Team Building: Enhances team bonding, offers immersive experiences, and creates virtual environments for collaborative activities. Eg. VR platforms like AltspaceVR and Engage VR provide virtual spaces for team meetings and social interaction. Top of Form

12.4.6 Environmental Monitoring and Management

Environmental monitoring utilizes technology to track and manage diverse aspects of the environment, involving the collection, analysis, and interpretation of data for understanding and responding to environmental changes. Examples include air and water quality monitoring systems, remote sensing satellites, smart grids for energy efficiency, wildlife tracking systems, and noise pollution monitoring devices. Some other examples are:

- The application of technology in environmental monitoring aims to oversee and regulate different facets of the environment by gathering, analysing, and interpreting data to comprehend and address environmental shifts.
- Systems measuring air quality employ sensors to detect pollutants, delivering real-time air quality data for swift intervention in worsening conditions.
- Devices monitoring water quality, equipped with sensors, ensure the safety of water bodies by identifying pollutants.
- Satellites with sensors capture data on land use, deforestation, and vegetation changes, facilitating large-scale environmental monitoring.

- Smart grids integrate technology for real-time monitoring and management of energy consumption, boosting distribution efficiency and reducing environmental impact.
- Wildlife tracking systems utilize GPS and sensors to monitor wildlife movements, aiding conservation efforts and ecological understanding.
- Devices measuring noise levels in urban areas identify sources of noise pollution, enabling the implementation of mitigation measures.

12.4.7 Precision Agriculture

Precision Agriculture, also referred to as precision farming or precision ag, utilizes advanced technologies to optimize farming practices. It employs information technology, sensors, and data analytics to enhance efficiency, productivity, and sustainability in agriculture. The primary goal is to enable precise decision-making in crop management, resource utilization, and overall farm operations.

Examples of Precision Agriculture Technologies:

- **Satellite Imagery:** Uses satellite data to monitor crop health, identify potential issues, and assess overall field conditions, aiding farmers in making data-driven decisions.
- **GPS-guided Equipment:** Implements GPS technology for accurate navigation and control of farm machinery during planting, fertilizing, and harvesting, minimizing resource wastage.
- **Sensor Technology:** Utilizes sensors to collect real-time data on soil moisture, nutrient levels, and crop health. The data is analyzed to optimize irrigation, fertilization, and pest control strategies.
- **Drones:** Deployed for aerial crop monitoring, providing high-resolution imagery. Drones facilitate quick and detailed assessments of large agricultural areas, aiding early problem identification.
- **Automated Machinery:** Incorporates automated and robotic machinery for tasks like planting, weeding, and harvesting. This reduces labour requirements and enhances precision in field operations.

12.4.8 E-Waste Management

E-waste management involves properly handling and disposing of electronic waste, such as computers and mobile phones, to mitigate environmental and health risks. The process includes organized collection from various sources, safe transportation to processing facilities, sorting, recycling valuable materials, and proper disposal of hazardous components. Secure data destruction is essential for data security before recycling.

Challenges include informal recycling practices in developing countries, the impact of rapid technological advances on recycling infrastructure, and a lack of awareness among consumers. Effective e-waste management offers benefits like resource recovery, environmental protection, energy conservation, and job creation in formal management systems.

E-waste management is vital for addressing environmental and health challenges linked to electronic waste. Practices like proper collection, recycling, and awareness contribute to resource conservation and environmental sustainability.

12.4.9 Climate Modelling and Prediction

Climate modelling and prediction encompass the utilization of sophisticated computational models to replicate Earth's climate system and forecast forthcoming climatic conditions. These models amalgamate diverse elements such as atmospheric, oceanic, and land processes while considering anthropogenic influences. They are essential for comprehending climate dynamics, evaluating potential impacts, and guiding strategies for mitigating and adapting to climate change.

Its essential components are:

Replication of Climate Processes: Climate models replicate intricate interactions among the atmosphere, oceans, land surfaces, and ice, capturing the dynamics of climate processes.

Anthropogenic Influences: Models factor in human activities like greenhouse gas emissions and land-use changes, significantly influencing the climate.

Forecasting and Anticipating: Climate models offer forecasts of future climate scenarios, aiding scientists and policymakers in anticipating changes and planning accordingly.

Analysis of Uncertainty and Sensitivity: Scientists perform analyses to comprehend the limitations and reliability of model predictions through uncertainty and sensitivity assessments, considering various input parameters.

Global and Regional Modelling: Models operate on global and regional scales, providing insights into broader climatic trends and localized impacts.

Verification and Validation: Model outputs undergo verification and validation against observational data to enhance accuracy and reliability.

Climate modelling faces significant challenges, including the complexity of addressing numerous interconnected factors, demanding substantial computational resources. Uncertainties within models arise from factors such as incomplete data, limited understanding of specific processes, and the intricate nature of Earth's climate system. Predicting regional impacts remains challenging due to the complexities of local climatic variations, and

ongoing efforts focus on enhancing the spatial and temporal resolution for more detailed and accurate predictions.

Applications: Climate models are crucial guides in policy development related to climate change mitigation and adaptation. They play a vital role in planning resilient infrastructure capable of withstanding future climate-related challenges. Understanding climate impacts contributes significantly to effective ecosystem management, aiding in biodiversity conservation and natural resource preservation. Climate modelling is instrumental in assessing risks associated with extreme weather events, sea-level rise, and various other climate-related phenomena.

Climate modelling and prediction are pivotal in comprehending, anticipating, and addressing Earth's climate system intricacies, thereby facilitating sustainable decision-making and adaptation efforts.

12.5 ROLE OF GOVERNMENTS, POLICY AND REGULATIONS

Governments play a pivotal role in addressing environmental challenges through legislative and regulatory measures. They institute laws covering emission standards, waste disposal guidelines, and conservation measures, establishing a legal foundation for sustainable practices and penalizing harmful activities. The formulation of proactive policies delineates the government's stance on environmental sustainability, guiding decision-making across sectors like energy, transportation, waste management, and natural resource conservation.

Incentive programs are deployed to encourage eco-friendly practices among businesses and individuals. These incentives, such as tax credits and subsidies, aim to promote sustainability, especially in renewable energy projects and energy-efficient technologies. Recognizing the global nature of environmental issues, governments engage in international cooperation, participating in agreements focused on climate change, biodiversity conservation, and pollution control.

Governments are responsible for monitoring and enforcing compliance with environmental regulations, employing robust mechanisms like inspections and penalties to ensure adherence. They also play a crucial role in raising public awareness through education campaigns, informing the public about the importance of conservation, recycling, and reducing carbon footprints.

Additionally, governments invest in research and development initiatives to advance sustainable technologies, providing funding and support for innovations in renewable energy, eco-friendly materials, and sustainable agriculture. In summary, governments not only establish regulatory frameworks but also incentivize, monitor, and enforce practices aligned with ecological well-being. Their multifaceted role extends to raising public awareness and supporting research for a more sustainable future.

Some of the ICT initiatives taken by the Indian government. National Digital Library (NDL), E PG Pathshala. e-Adhyayan, Shodhganaga platform, e-ShodhSindhu platform, e-yantra, FOSSEE((Free/Libre and Open Source Software for Education), Spoken Tutorial portal, Digi Locker, PMJDY, PAHAL

12.6 ROLE OF CORPORATE SECTOR IN ACHIEVING SUSTAINABLE DEVELOPMENT THROUGH ICT

The corporate sector drives sustainable development by integrating Information and Communication Technology (ICT). Key aspects include:

- **Innovation and Technology Adoption:** Corporations innovate with smart technologies, IoT, and data analytics for resource management and environmental monitoring.
- **Operational Efficiency:** ICT enables streamlined processes, data-driven decisions, and lower resource consumption, enhancing operational efficiency.
- **Supply Chain Sustainability:** Through ICT like blockchain, corporations ensure transparent and sustainable supply chains, promoting ethical sourcing and reducing environmental impact.
- **Renewable Energy Integration:** Many invest in renewable energy solutions and use ICT for smart grids, energy management, and real-time monitoring.
- **Remote Work and Telecommuting:** ICT tools facilitate remote work, reducing commuting, and carbon emissions, and enhancing work-life balance.
- **Environmental Monitoring and Reporting:** ICT enables corporations to monitor and report environmental impact, aiding in continuous improvement strategies.
- **Data-driven Sustainability Reporting:** Corporations use ICT to collect, analyze, and report sustainability data, promoting accountability and responsible business behaviour.
- **Eco-friendly Product Design:** ICT supports eco-friendly product design by optimizing materials and manufacturing processes.
- **Corporate Social Responsibility:** ICT enhances CSR initiatives through community support, education investment, and technology-driven social impact projects.
- **Collaboration and Partnerships:** Corporations collaborate within and beyond industries, leveraging ICT for communication, knowledge-sharing, and impactful sustainability initiatives.

Examples:

- Tesla utilizes ICT for smart grid solutions, optimizing energy consumption in electric vehicles.
- IKEA uses blockchain for supply chain transparency, ensuring ethical sourcing of materials.
- Google employs remote work tools like Google Workspace, reducing carbon emissions from commuting.
- Apple employs data analytics to monitor carbon emissions across its supply chain, informing sustainability strategies.
- Unilever leverages ICT for sustainable product design, incorporating recycled materials in packaging.

12.7 UNDERSTANDING INTERNATIONAL FRAMEWORKS RELATED TO ICT AND ENVIRONMENTAL SUSTAINABILITY

International frameworks in Information and Communication Technology (ICT) and environmental sustainability aim to guide global efforts towards a more eco-friendly and responsible use of technology. Key frameworks include:

United Nations Sustainable Development Goals (SDGs):

- SDGs encompass environmental sustainability goals, recognizing ICT as an enabler.

SDG 7 emphasizes affordable and clean energy, utilizing ICT for smart grids and energy-efficient technologies.

ITU-T Recommendations:

- ITU provides recommendations for using ICT to achieve environmental sustainability.

It recommends methodologies for assessing the environmental impact of ICT.

ISO 14001 - Environmental Management Systems:

- ISO 14001 sets criteria for environmental management systems.

Organizations use ISO 14001 to manage environmental aspects, including those related to ICT.

ICLEI - Local Governments for Sustainability:

- ICLEI guides local governments for sustainable development, integrating ICT.

Cities use ICT in smart city initiatives to optimize resource use and reduce environmental impact.

Green ICT Initiative by GeSI:

- GeSI promotes ICT for environmental and social sustainability.
- GeSI's Green ICT toolkit aids companies in assessing and improving the sustainability of their ICT practices.

The Global e-Sustainability Initiative (GeSI):

GeSI works on initiatives highlighting the role of ICT in sustainable development.

- Example: GeSI collaborates on tools like the "Digital with Purpose" report, emphasizing ICT's positive impact on sustainability.

Climate Savers Computing Initiative:

- Focuses on improving energy efficiency in computers and servers.

Companies like Google and Intel commit to reducing ICT-related carbon emissions.

Circular Economy Action Plan by the European Union:

Aims to make products more sustainable and reduce waste, emphasizing ICT.

Implementation of circular economy practices in designing and disposing of electronic devices.

These frameworks offer guidelines, standards, and initiatives for aligning ICT practices with global environmental sustainability goals. Organizations and governments can leverage these frameworks to shape policies, strategies, and actions for a more sustainable use of ICT.

12.8 CASE STUDIES AND BEST PRACTICES

Few Best Practices and Case Studies in the Field of ICT and Innovation.

1) Aadhaar (India):

- *Aadhaar, India's biometric identification system, is a noteworthy ICT initiative, providing a unique identity to residents for efficient service delivery and reducing fraud.*
- *The Aadhaar system streamlined government services, improved subsidy distribution, and enhanced financial inclusion.*

2) Aarogya Setu App (India):

- *The Aarogya Setu app, launched by the Indian government, played a crucial role in contact tracing and providing COVID-19-related information.*

- *It showcased the use of technology for public health initiatives during the pandemic.*

3) UPI (India):

- *Unified Payments Interface (UPI) revolutionized digital payments in India, enabling seamless, real-time transactions between banks through mobile devices.*
- *UPI's success led to a surge in digital transactions, promoting a cashless economy and fostering financial inclusion.*

4) Bharti Airtel's Mobile Services (India):

- *Airtel's mobile services played a pivotal role in connecting people across India, leveraging technological advancements to offer reliable and affordable mobile communication.*
- *Airtel's growth and innovative services showcase the impact of ICT in expanding connectivity and bridging communication gaps.*

5) Smart Cities Mission (India):

- *India's Smart Cities Mission focuses on using ICT for urban development, aiming to enhance infrastructure, improve services, and promote sustainability in selected cities.*
- *Projects in cities like Pune and Bhubaneswar demonstrate the implementation of smart technologies for efficient waste management, traffic control, and citizen services.*

6) E-Government in Estonia:

- *Estonia excelled in e-government, providing digital public services and e-residency options, showcasing streamlined governance through digital platforms.*
- *Estonia's digital services, including e-residency and e-Governance, illustrate how ICT enhances government efficiency and citizen engagement.*

7) Silicon Valley Tech Innovation:

- *Silicon Valley in the United States is a global hub for tech innovation, with companies like Apple, Google, and Facebook continuously innovating and transforming the use of ICT.*
- *The success stories of Silicon Valley companies highlight the impact of fostering an innovation ecosystem, attracting talent, and driving economic growth.*

8) South Korea's 5G Deployment:

- *South Korea has been a leader in 5G deployment, transforming communication with ultra-fast internet speeds and low latency.*
- *The rapid adoption of 5G in South Korea showcases the potential of ICT in revolutionizing connectivity and enabling emerging technologies.*

9) M-Pesa in Kenya:

- *M-Pesa, a mobile money service in Kenya, transformed financial transactions, especially in rural areas, by leveraging ICT.*
- *M-Pesa's success demonstrates the impact of innovative mobile solutions in enhancing financial inclusion and accessibility.*

These case studies exemplify how ICT and innovation contribute to societal development, economic growth, and improved services, providing valuable examples for sustainable development globally.

Check Your Progress: 3

Note:1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of this unit

1. Explain why these best practices are considered exemplary and how they contribute to the overall efficiency, sustainability, or innovation within the society.

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12.9 INNOVATION AND ENTREPRENEURSHIP IN DEVELOPING ICT SOLUTIONS FOR ENVIRONMENTAL SUSTAINABILITY

Technology innovation continuously propels the creation of eco-friendly ICT solutions. Entrepreneurs, recognizing opportunities in emerging technologies, develop sustainable products and services to address environmental challenges.

- **Renewable Energy Integration:** Innovative ICT solutions emerge, integrating renewable energy sources to enhance sustainability in power consumption. Entrepreneurial startups focus on providing renewable energy solutions for data centres or promoting energy-efficient devices.

- **Smart Infrastructure:**Innovation drives the creation of smart grids, intelligent transportation systems, and connected infrastructure, optimizing resource use. Entrepreneurs establish companies specializing in developing and implementing smart city solutions for enhanced sustainability.
- **IoT and Environmental Monitoring:**Leveraging the Internet of Things (IoT), innovation enables real-time environmental monitoring and data collection. Entrepreneurs, in response, develop IoT devices and platforms to monitor air quality, water usage, and various environmental parameters.
- **Circular Economy Solutions:**Innovative design focuses on ICT products aligned with the circular economy, emphasizing reuse and recycling. Entrepreneurs launch businesses refurbishing electronic devices or promoting sustainable e-waste management.
- **Data Analytics for Sustainability:**Innovation employs data analytics to optimize resource use, reduce waste, and enhance overall sustainability. Entrepreneurs develop analytics platforms for businesses to measure and improve their environmental impact.
- **Collaboration Platforms:**Innovation leads to the creation of collaborative platforms, enabling organizations to work together on sustainability initiatives. Entrepreneurs establish businesses facilitating collaboration for shared sustainability goals.
- **Educational and Awareness Tools:**produces tools and technologies for environmental education and awareness. Entrepreneurs create platforms engaging users in sustainable practices and providing information on eco-friendly choices.
- **Remote Work Solutions:**Innovation supports the development of technologies for remote work, reducing the need for commuting and associated environmental impact. Entrepreneurs offer solutions for virtual collaboration, decreasing the ecological footprint of traditional office-based work.
- **Policy and Advocacy Platforms:**Innovation creates platforms using ICT to influence and support environmental policies and advocacy. Entrepreneurs establish businesses focused on influencing policymakers and promoting sustainable practices.

12.10 ETHICS AND SOCIAL RESPONSIBILITY

Ethics in environmental sustainability is crucial for creating a world where human activities harmonize with nature, fostering a healthy and resilient planet for current and future generations. It involves a commitment to ethical decision-making, accountability, and a shared responsibility for the well-being of the Earth and its ecosystems.

1. **Stewardship:** Acknowledging a responsibility to care for and protect the environment.
2. **Responsibility:** Recognizing the impact of human activities on the environment and taking responsibility for minimizing negative effects.
3. **Justice and Equity:** Ensuring fair distribution of environmental benefits and burdens. Addressing environmental issues with consideration for vulnerable and marginalized communities.
4. **Intergenerational Equity:** Considering the needs and rights of future generations in current decision-making.
5. **Transparency:** Providing clear and accurate information about environmental practices.
6. **Conservation and Biodiversity:** Supporting conservation efforts to protect biodiversity. Avoiding actions that contribute to the loss of species or ecosystems.
7. **Corporate Social Responsibility (CSR):** Integrating ethical environmental practices into business operations. Balancing economic interests with social and environmental responsibilities.
8. **Community Engagement:** Involving communities in decision-making processes related to environmental sustainability.
9. **Global Cooperation:** Recognizing that environmental challenges often transcend national borders. Promoting international collaboration to address global sustainability issues.
10. **Ethical Consumption:** Encouraging individuals to make ethical choices in their consumption patterns. Supporting environmentally friendly products and practices.

Check Your Progress: 4

Note: 1) Use the space provided below for your answers.

2) Compare your answers with those given at the end of this unit

1. Choose a recent innovation in Information and Communication Technology (ICT) that has significantly impacted society

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12.11 LET US SUM UP

“Technology is neither good nor bad; nor is it neutral. It is the use that makes it good or bad.” Melvin Kranzberg.

So, in conclusion, we can say that the integration of Information and Communication Technology (ICT) with sustainability initiatives is imperative for addressing environmental challenges and fostering a more resilient and eco-friendly future. As evidenced by best practices and case studies, ICT innovations play a pivotal role in streamlining services, promoting financial inclusion, enhancing connectivity, and contributing to urban development. International frameworks guide aligning ICT practices with global sustainability goals. The COVID-19 pandemic has further underscored the significance of digital solutions in crisis response and management.

12.12 KEYWORDS

Green data centre : A green data centre is a facility that stores, manages, and distributes data while minimizing environmental impact and maximizing energy efficiency

Carbon footprint : A carbon footprint is a value that compares the amount of greenhouse gases that a person, product, company, or country releases into the atmosphere. It's a way to measure the impact of an activity.

Digital twin : A digital twin is a digital model of an intended or actual real-world physical product, system, or process that serves as the effectively indistinguishable digital counterpart of it for practical purposes, such as simulation, integration, testing, monitoring, and maintenance.

Blockchain : is a method of recording information that makes it impossible or difficult for the system to be changed, hacked, or manipulated. Blockchain technology is applied to supply chains to ensure transparency and traceability, especially in industries like food, where consumers can trace the origin of products.

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12.14 CHECK YOUR PROGRESS: POSSIBLE ANSWERS

Check Your Progress: 1

1. Give real-life examples on the following:
 - a) Energy Efficiency
 - b) Waste Reduction
 - c) Sustainable Practices
 - d) Alignment with Environmental Sustainability Goals
 - e) Carbon Footprint Reduction
 - f) Resource Conservation
 - g) Community Engagement
 - h) Long-Term Environmental Impact
 - i) SustainableTechConnect

Check Your Progress: 2

1. ICT acts as a catalyst for inclusive and equitable access to education, healthcare, and essential services in developing regions. By overcoming geographical barriers, fostering connectivity, and providing digital solutions, ICT contributes significantly to sustainable development goals by ensuring that everyone, regardless of their location, has access to vital services and opportunities.

Check Your Progress: 3

1. For example, Aarogya Setu's best practices, including efficient contact tracing, sustainability in pandemic management, innovation in digital health solutions, privacy measures, community engagement, and integration with public health initiatives, collectively make it exemplary. The app demonstrates how ICT innovations can effectively contribute to public health, especially during a crisis, while addressing concerns related to privacy and fostering community participation.

Check Your Progress: 4

1. The recent innovation in Information and Communication Technology (ICT) that has significantly impacted society is the widespread adoption and advancement of 5G technology.

5G provides ultra-fast internet connectivity, significantly enhancing download and upload speeds compared to its predecessors. This has transformed the way people access and share information, fostering a more connected society. 5G technology has had a profound impact on society, influencing connectivity, technological advancements, healthcare, the economy, and global interactions. Its widespread adoption marks a significant milestone in the evolution of ICT, shaping the way people live, work, and communicate in the digital age.

UNIT 13 IMPLEMENTATION OF ICT FOR DEVELOPMENT (4D)

Structure

- 13.0 Introduction
- 13.1 Learning Outcomes
- 13.2 Introduction to ICT for Development
 - 13.2.1 Definition of ICT for Development and its Significance in Global Development.
 - 13.2.2 Brief Historical Overview of the Evolution of ICT for Development Initiatives.
- 13.3 Key Components of ICT for Development
 - 13.3.1 Digital Infrastructure: Importance of Reliable and Accessible Digital Infrastructure.
 - 13.3.2 Human Capacity: Role of Education and Training in Building Human Capacity for ICT for Development.
- 13.4 Case Studies of Successful ICT for Development Projects
 - 13.4.1 Highlights of Specific Projects that have Effectively Implemented ICT for Development.
- 13.5 Challenges in ICT for Development Implementation
 - 13.5.1 Digital Divide: Analyzing Disparities in Access to Technology and Information.
 - 13.5.2 Cultural and Societal Factors: Discussing How Cultural Nuances Affect the Acceptance and Success of ICT for Development Initiatives.
- 13.6 Ethical Considerations in ICT for Development
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 - 13.6.2 Ensuring Inclusivity: Strategies to Prevent Exclusion and Ensure the Benefits of ICT for Development reach all Segments of Society.
- 13.7 Let Us Sum Up
- 13.8 Keywords
- 13.9 Further Readings
- 13.10 Check Your Progress: Possible Answer

13.0 INTRODUCTION

Debate around development started in 1950s and 1960s and continue in the present times. From defining development to specifying the path to development, there has not been an absolute answer so far. But certainly there are some issues which have been identified to be integral to development. For instance, eradicating poverty, eliminating illiteracy and

ensuring physical as well as mental health have become intrinsic to development.

To achieve these targets, many tools have been used for purpose of development all over the world. Information, Communication and Technology (ICT) too have emerged as an important tool for development. ICT largely refers to use of digital technologies in any sector and since, these ICTs are being used for development, this use of ICTS in the development process is termed as ICT for Development.

ICT for Development to a large extent has benefited the process of development as right from the modernization or dominant paradigm of development, use of technology has been encouraged. Many initiatives have been taken for development in different areas including health, agriculture and education, where ICTs are being used to benefit the communities. Though we need to note at the same time that the integration of ICTs has not remained easy and smooth as both the physical and human resource infrastructure are still taking shape in some of the countries particularly the developing countries. We need to note that the implementation of ICT for Development has not been without challenges. Hence, this Unit would introduce you to the journey of evolution of ICT4D and the present situation especially in terms of challenges and ethical concerns.

13.1 LEARNING OUTCOMES

After reading this unit, you will be able to:

- understand the evolution of ICT for Development initiatives;
- comprehend the key components of ICT for Development;
- analyse the challenges and ethical concerns of ICT for Development; and
- gain an insight into the successful ICT for Development projects.

13.2 INTRODUCTION TO ICT FOR DEVELOPMENT

ICT4D refers to use of Information, Communication and Technology (ICT) for Development. This term became popular with the increasing use of ICTs in different sectors for development all over the world. As we have learnt in the history of development paradigms that technology was taken to be an important component of development right from the initial definitions given under the modernization paradigm, the ICTs nowadays have become an integral part of development. Use of ICTs can be traced back to Daniel Lerner's belief in the potential of technologies, largely mass media, to modernise traditional societies (Melkote & Steeves, 2015). Hence, use of ICT for Development is not a recent approach to development but has evolved over a period of time from 1950s onwards. Therefore, ICTs do not only include the digital technologies of the present times but also comprises of the

analogue technologies of radio and television (Melkote & Steeves, 2015). This integration of ICTs has happened owing to adoption of ICTs in every aspect of lives in all the countries of the world including the developing countries. So, before going further, let's look into the definitions of ICT for Development and its evolution in detail.

13.2.1 Definition of ICT for Development and its Significance in Global Development

Heeks (2018) defined ICT for Development as “the application of any entity that processes or communicates digital data in order to deliver some part of the international development agenda in a developing country” (p.11). Melkote & Steeves (2015) states that ICT for Development can be referred to as possibilities created through access and use of ICTs for modernization.

UN (n.d.) says that the World Summit on the Information Society (WSIS) Geneva 2003 – Tunis 2005, the Copenhagen Declaration and Tunis Commitment all emphasized on the potential of ICTs in the overall development of the countries including economic, political and social development. Further recognizing the role of ICTs in development, UN mentions that ICTs enable individuals and communities to create, access, utilize and share information and knowledge to achieve full potential in promoting their development and improving their quality of life.

ICT for Development is becoming increasingly popular as the integration of ICTs has made the process of development cheaper (relative cost- one computer can serve many), quicker and benefits many at the same time. But at the same time, it should be noted that as ICT for Development is a concept generally applicable to developing countries, the challenges in the developing countries are still prevalent in terms of less resources, inequality and less investment. Therefore, there is no denying the fact that implementation of ICT for Development is important not only for the developing countries but also for the development at the global level but the implementation needs to be done, taking challenges in its ambit.

13.2.2 Brief Historical Overview of The Evolution of ICT for Development Initiatives

As you have learnt in earlier Units that from the beginning of discussions around development that technology has been an important pillar of development, ICT for Development has extended this aspect of technology as it emphasized on the use of Information and Communication Technology in all aspects of lives rather than being limited to use of technology in the production process. The term ICT evolved from IT (Information Technology) to include Communication aspect. The term IT was much limited in nature as it intends to provide only information to its users rather than communicating with them. IT was more to do with government, multinational corporations and one-dimensional economic growth rather than over-all development. Inclusion of ‘Communication’ in IT made it more encompassing as ICTs aimed to not only give information to its users but also to give them a

platform to share and communicate with the provider as well as fellow users. Further, this process of sharing information and ideas with the provider of information (which remains most of the times, government or sometimes non-governmental organisations as well) and fellow users was gradual and not sudden. Rather ICT in the beginning was almost equivalent to IT and hence, can be called ICT for Development 0.0, in which communication or interaction aspect was lacking. Heeks (2008) mentions that ICT for Development 0.0 was chiefly used in government offices to share information inside offices only whereas in ICT 1.0, ICTs were brought out from confines of government offices to rural areas. Most popular projects under ICT 1.0 in the first decade of 21st century were tele-centres where a computer or desktop was installed with an internet connection in a small room which largely was meant to provide information to the villagers (Heeks, 2008). The idea for these projects were directly taken from developed western world and implemented in rural areas of developing world without much consideration of socio-cultural context of developing countries. Hence, these projects did introduce technology in these far-off areas but could not prove much beneficial and sustainable in nature as community involvement would not be much generated. This was also the time when eight Millennium Development Goals (MDGs) were adopted by UN at the international level to define the direction of development in 2000. A period of 15 years was kept to achieve these goals and though the role of ICTs was not explicitly explained but ICTs was taken as one of the important tools to achieve development under MDGs since some of the ICT for Development were already showing results. After these initial projects, came the projects under ICT for Development 2.0 where interaction and feedback was given more importance to make the projects more sustainable. Heeks (2008) says that there is no clear demarcation to tell when ICT for Development 1.0 ended and ICT for Development 2.0 started rather, ICT 2.0 is still evolving. This phase may be related to adoption of 17 Sustainable Development Goals (SDGs) after the experience of MDGs in which the role of ICTs was made an implicit part of some of the targets under specific goals. Figure 13.1 shows the evolution of ICT for Development and perceptions related to it as to how ICTs were ignored or isolated in the beginning, slowly giving way to integration and innovation in ICT for Development projects.

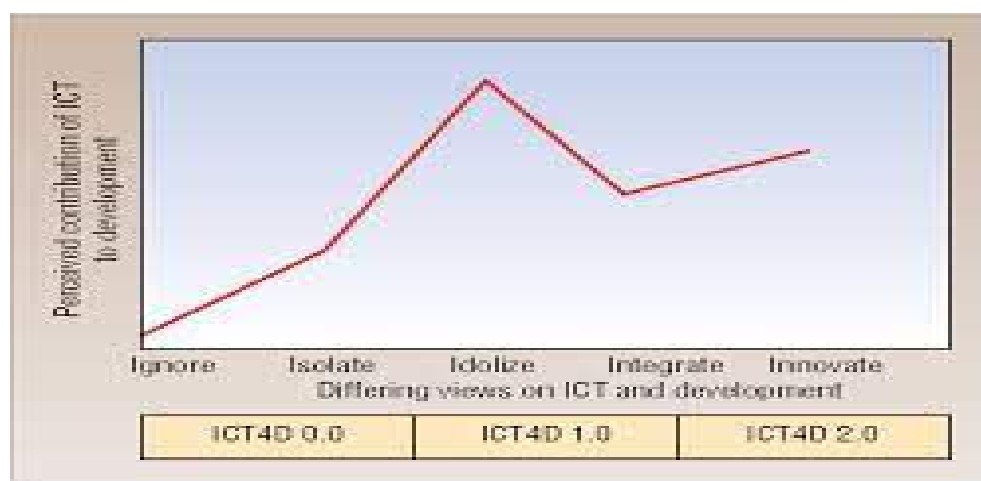


Figure 13.1: Changing Views on ICTs and Development (Source: Heeks (2008))

Check your Progress: 1

Note: 1) Use the space below for your answer.

2) Compare your answers with those ones given at the end of this unit.

1. What do you understand by ICT for Development?

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13.3 KEY COMPONENTS OF ICT FOR DEVELOPMENT

ICT for Development comprises of two key components namely, establishment of Digital Infrastructure and building of Human Capacity. Both are equally important as one cannot function without another. In fact, some of the early initiatives have not been very successful as more emphasis was made on one aspect of the ICT for Development ignoring the other component. Hence, we need to learn both the components in detail which are as follows:

13.3.1 Digital Infrastructure: Importance of Reliable and Accessible Digital Infrastructure

Digital Infrastructure refers to the establishment of physical infrastructure needed for development in different sectors of the economy and society. Largely, it consists of technologies which are needed to reach the participants of the development process. Among other technologies, it may include radio, television, mobile phones, computers and internet connectivity. We have seen that these technologies have been made available in many parts of the world including the developing world and the farther rural areas. Nearly 70% of the world population has become internet users (ITU, 2023). But most of the infrastructure had not been sustainable in its working. Most of the development projects have been made operational one-time but not much attention had been given to make it sustainable. Problems like irregular supply of electricity and repair and maintenance have lead to discontinuance of these projects. But lately, with the reach of handy digital devices and widespread network of internet, digital infrastructure is gaining foothold in the not-so accessible areas.

Accessibility of digital infrastructure also needs to be understood in terms of language issues. As we know that most of the technologies have been developed by the multinational corporations which use English as their only language, people who do not understand English face a lot of challenges in accessing these technologies. Though many initiatives have now incorporated regional languages in their digital content, but still more attention is needed in understanding this aspect of accessibility of ICTs.

13.3.2 Human Capacity: Role of Education and Training in Building Human Capacity for ICT for Development

Human capacity refers to the ability and capacity of the people to operate the digital infrastructure. The aim under building human capacity remains not only to just operate the technologies but to operate them in such a manner that maximum output may be derived from the use of technology for the community around. Establishing the digital infrastructure is just one part of the ICT for Development initiatives, inculcating digital skills among the community is equally important part of these initiatives to utilize the full potential of these technologies. Imparting skills and building capacities among community members to operate these ICTs are also important for sustainability of these digital initiatives as some of the early projects failed owing to lack of human resources who could operate and utilise these technologies. It happened that when the physical infrastructure was set-up, some of the experts came to operate it but left after sometime leaving the infrastructure useless as nobody in the community knew how to maintain and utilise the infrastructure.

Digital skills are the skills which are needed to operate the digital infrastructure. These also include the language skills as most of the content available through ICTs is in English language which might not be understood at regional and local levels especially in rural areas.

Check your Progress: 2

Note: 1) Use the space below for your answer.

2) Compare your answers with those ones given at the end of this unit.

1. List the key components of ICT for Development.

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13.4 CASE STUDIES OF SUCCESSFUL ICT FOR DEVELOPMENT PROJECTS

Numerous projects related to ICT for Development have been undertaken in different parts of the world starting from early 1970s to the present times. As we have discussed earlier, ICT for Development projects not only utilised the recent digital technologies but also mass media like radio and television to reach the marginalised sections in different countries. These projects have been implemented in various areas like agriculture, education, health and small-rural entrepreneurship. In the next section, we would discuss some of the projects which may be termed as successful stories of ICT for Development projects. It may be kept in mind that these projects might not be able to achieve all the targets that they had set for the beneficiaries but were

at least able to utilize comparatively new tools of development, ICTs, in the development process and could learn from the lacunae that hindered the full potential of the projects.

13.4.1 Highlights of Specific Projects That have Effectively Implemented ICT for Development

The Satellite Instructional Television Experiment (SITE) in India, the Palapa Experiment in Indonesia and experiments with satellite-based rural telephony in Peru were some of the earlier examples of ICTs in the 1970s and 1980s (Melkote & Steeves, 2015).

SITE was a one-year experiment conducted during 1975-76 in India in which satellite Television was used for disseminating information related to various aspects including agriculture, education and health for diverse rural population. Nearly 2400 villages were covered under the project belonging to different linguistic, cultural, climatic and agricultural regions of the country. Programme content was also a combination of centralization and decentralization as some of the content was produced in centrally located studios with the help of experts while some of the programmes were produced at local level to connect to the local population. SITE was designed to give policymakers, programme producers or social scientists insights into the different ways in which TV can be used to further development in diverse socio-economic and cultural situations.

Palapa Ring project in Indonesia initiated in 1976 aimed at connecting all the islands in the country. The project attempted to provide internet access to all provinces in the country by 2019 but the evaluation of the project shows that internet access provided under the project has led to increase only in personal computing consumption whereas the processes at workplaces or educational institutions have not benefited much from it (Gupta, n.d.). The internet access under the project has reached the villages but information being provided to villagers need to be more specific to their needs to reap maximum benefit from the use of ICTs in development projects.

Rural Communication Services Project (RCSP) was launched in Peru to integrate satellite communication in rural development. The project was based on the premise that connected villages through telecommunications would be able to overcome infrastructure and other resource limitations. The project was able to introduce telephony in many rural areas of Peru but strong institutional support, sound management and dynamic leadership was needed to make it fully beneficial to the users (Tietjen, Shaw & Block, 1987).

Another popular project launched in India, which in a way was among the pioneers in e-governance project was Gyandoot. Gyandoot, which means “Purveyor of Knowledge” in Hindi, was an intranet-based service portal, implemented in the Dhar district of the state of Madhya Pradesh in India, in 2000 (Bhatnagar et al, 2003). This portal was made available to villagers through information kiosks which were set-up in different villages. Villagers

with the help of a person who was managing the kiosk were able to register their complaints or grievances to government officials sitting in their villages and can also access information related to different fields including crop prices or details of land. The evaluation of the project showed that people felt that they were able to approach government officials much quicker through these kiosks (Bhatnagar et al., 2003).

Bringing in e-governance, governments of many countries have started their own web portals to provide information related to all services and schemes of the government at one platform. National portal of India (NIC,n.d.) was one such initiative taken up by Indian government in 2005 and continue to provide all types of information related to varied fields like agriculture, social development, law & justice, environment or youth & sports.

With the increasing penetration of digital technologies including mobile phones, computers and internet after 2000s, many governmental and non-governmental organisations have introduced the projects related to these digital technologies where various services were provided to users on their mobile-phone handsets. Infact, many people belonging to marginalised sections especially in rural areas were provided with digital devices like mobile phones and laptops to access the benefits of ICTs in the development process. But again here, challenges like lack of maintenance facilities and lack of digital skills have slowed down the progress of these projects to some extent. However, after the evaluation of these projects, more sustainable strategies have been adopted where community members have been involved in ICT4D right from the decision-making to implementation and finally evaluation of projects.

Activity 1

Identify any ICT for Development in your nearby area and analyse its implementation.

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13.5 CHALLENGES IN ICT FOR DEVELOPMENT IMPLEMENTATION

Implementation of ICT4D has not been without challenges. Two challenges namely digital divide and socio-cultural differences have remained primary in the use of ICTs for Development.

13.5.1 Digital Divide: Analyzing Disparities in Access to Technology and Information

Digital Divide as the name suggests refers to the gap that has developed owing to the lack of access and use of digital technologies among some countries. Largely, the developing world lacked access and use of digital

technologies hence, adding up one more parameter of lagging behind the developed world or the Global North. A report by ITU (2017) found that European countries continue to lead in the development of ICTs whereas countries in the continent of Africa have still seen the least development of ICTs. It should be noted here that it is not only the reach and access that has created disparities in the use of ICTs but also the skill-level of users that has lead to divide called the digital divide. Infact, most of the times the focus remains on making a digital technology accessible among community members by the government as well as non-governmental organisations while paying no attention to make the community members capable of using the technology. Rather in the early initiatives, organisations introducing the technology used to bring in their own expert personnel to operate the technologies, making the community members dependent on this person. But with coming up of personal digital devices like mobile phones, digital literacy has become an important part of implementation of ICT for Development.

A closer look would reveal that this digital divide not only exists in terms of countries but many other sections of society face this digital divide. For instance, gender digital divide has also become very apparent as less number of women compared to men would usually be able to access and use digital technologies. A report by ITU (2017) mentions that digital gender divide is relatively small in developed countries, more pronounced in developing countries whereas it is substantial in Least Developed Countries (LDCs). Similarly, rural areas as compared to urban areas would face many barriers in accessing digital technologies, keeping rural areas or villages at a disadvantage. ITU (2023) states that only 50% of people in rural areas as compared to 81% in urban areas are able to access internet. Even in case of age-groups, we would see that generally youth would have both access and skills to use ICTs whereas elderly would remain aloof from the digital world. A report by ITU (2017) states that the proportion of population aged 15-24 years who are online is estimated to be over 70% as compared to 48 % of the overall population, adding that elderly people are less likely to be online.

13.5.2 Cultural and Societal Factors: Discussing How Cultural Nuances Affect the Acceptance and Success of ICT for Development Initiatives

Socio-cultural factors have remained another big challenge in the implementation of ICTs. As we have seen in the evolution of ICT for Development, most of the digital initiatives have been taken up originally in the developed world of Western countries and replicated in developing world of Eastern countries. Under modernisation or dominant paradigm of development, when adoption of technology was thought to be one of the major paths to be taken for development in developing countries, these digital initiatives were implemented in the developing countries without any consideration of socio-cultural factors. Rather in the 1950s and 1960s, a clear demarcation was made in terms of characteristics of traditional and modern societies. Along with more use of technology in modern societies as compared to traditional societies, traditional societies were also termed as

having less empathy, close-knit communities, limited view of the world and less open to innovation and experimentation. Hence, all these factors played a role in hindering the adoption of digital technologies and still continue to slow down the rate of adoption of ICTs in the developing countries. For instance, community members in the developing countries believe in subsistence production, the production which is able to fulfil their own needs and they think that if they aspire for more, they might interfere with the share of others. Therefore, they still would not appreciate the technologies which encourage mass production. On the similar lines, religion has always remained a very important part of their lives and hence, use of many technologies at times do not go by their beliefs and norms. Gender too plays an important role in many countries. Women are still not considered to be capable enough to use ICTs independently. ITU (2023) states that there is only 5% difference in usage of internet among men and women at the world level but increases to 10% in case of Africa. Hence, socio-cultural factors of every community needs to be kept in mind before introducing a technology so that the technology can be used efficiently and effectively by the community members.

Figure 13.2 depicts the changing issues and concerns regarding the use of ICT for Development over time as it shows that the implementation of ICT for Development starts with reach and accessibility of ICTs in a particular community and ends with the culminates with the evaluation of goals of the community.

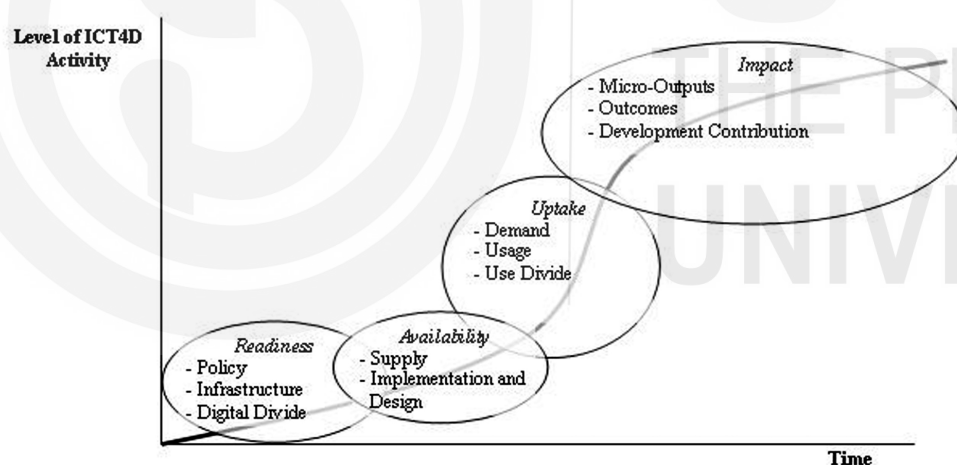


Figure 13.2 : Changing ICT for Development Issues Over Time (Source: Heeks (2008))

Check your Progress: 3

Note: 1) Use the space below for your answer.

2) Compare your answers with those ones given at the end of this unit.

1. How do you think digital divide hinders ICT4D?

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Activity 2

1. Look for a project related to ICT for Development in your nearby area and analyse if implementing agency faced any challenge in implementing it.

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2. Identify if there is any particular group of community members who have not been able to access (digital divide) the ICT for Development.

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13.6 ETHICAL CONSIDERATIONS IN ICT4D

Use of ICTs have been much appreciated and accommodated despite of some of the challenges in most of the countries as the digital world seems to be the only way forward. But still there are some ethical issues which are being discussed in the present times regarding the use of ICTs for development. Invasion of privacy and ensuring inclusivity are being seen as concerns from ethical point of view in implementation of ICT for Development. Both the concerns are discussed below:-

13.6.1 Privacy Issues: Ethical Implications of Data Collection and Usage in Development Projects

ICTs are being seen as invasive technologies as most of the technologies are individualistic in nature. To make these technologies more effective and custom-made, these technologies tend to collect lot of personal information of the people whom it intend to benefit. But since people especially in the developing countries are not much used to sharing personal details like age and income, people in these countries become all the more scared of the use of these technologies. Furthermore, since many of the projects related to ICT4D especially in the initial period were handled by Multinational Corporations (MNCs) and many trace the origin of technology driven initiatives to western capitalistic world, thinkers in the field of ICT4D and users themselves remain wary of the motives of implementing agencies of ICT4D. Many a times, the data collected for implementation of ICT4D has been said to be used by MNCs for furthering their commercial interests. Rather some studies have pointed out that this data is also being used for political purposes too as this data is quite vast. Infact, a term has also been devised for this kind of data, Big data. Big data is the large datasets which is being generated with the help of ICTs and can be used to reach large number of people with specific aim and may also be used to analyse the attributes of the general public.

13.6.2 Ensuring Inclusivity: Strategies to Prevent Exclusion and Ensure the Benefits of ICT4D Reach all Segments of Society

As we have discussed above in the challenges to the implementation of ICT4D that digital divide has remained an important concern, ensuring inclusivity has also emerged as an ethical concern owing to the diversity that is found in different societies and communities. Infact, 'inclusive development' has become an important term in the development studies. It refers to development of all irrespective of gender, caste, class, religion or race. It aims to develop each and every member of the society or community rather than benefiting some members of the community. The debate around inclusivity started immediately after the modernization or dominant paradigm as the steps taken under this paradigm had led to development of only few sections of the society leaving many behind and hence, creating a further gap between 'haves' and 'have-nots'. But lately this has been realised that the strategies adopted in the early stages of development need to be revised to take everyone along. Taking everyone along would not only make the development an inclusive one, but also a sustainable one where we need not care about the future resources or generations as inclusive development would ensure stability. To bring in all the stakeholders together, various strategies are being deployed by the government as well as NGOs. To ensure inclusivity, horizontal approach to development and communication is being encouraged instead of vertical approach. Rather in the present times, both the approaches are combined to elicit maximum benefit. Horizontal approach to development and communication refers to bringing in all the stakeholders of the development process at one place, holding discussions and devising the development plan instead of implementing the plans made by central-level experts in a far-off community paying less attention to the specific needs of the community. Vertical approach emphasizes top-down approach whereas horizontal approach encourages bottom-ups strategy in which community members themselves are encouraged to identify their problems and devise their own path to development. Horizontal development also encourages self-development whose output seems to be more as when people own and take responsibility for their development, they work more passionately for it, hence, maximizing the benefit for all. Similarly, Participatory approaches have also been encouraged for undertaking development where community members participate in each and every step of development from decision-making to generating resources for the project to actually executing the project on the ground-level and then also evaluating the project. This evaluation of the project provides a road-map for future development projects.

Check your Progress: 4

Note: 1) Use the space below for your answer.

2) Compare your answers with those ones given at the end of this unit.

1. What strategies are being incorporated in ICT4D projects to ensure inclusivity?

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13.7 LET US SUM UP

ICT4D has become an integral part of the development process as ICTs have become intrinsic to all aspects of our lives. Having being initiated into the process of development in early 1990s as one of the tools, today ICTs have become an important tool as the reach and access of ICTs have increased mutli-fold. Many initiatives are being taken by keeping ICTs at the center both by government as well as non-government organisations. But the implementation of ICT4D has not been without challenges and concerns. Establishment of digital infrastructure and building human capacities are still posing themselves as challenges in the path of ICT4D. Also, sometimes intrusive use of ICTs and ensuring inclusivity too become ethical concerns owing to diversity of stakeholders of the development process. But despite all these challenges and concerns, initiatives in ICT4D are being taken up by development agencies and transversing on the development path by these agencies is also taking place at the same time with the active participation of community members.

13.8 KEYWORDS

Digital Divide

: It is the gap that exists between those who are able to access and use digital technologies and those who are not able to.

Digital Literacy

: It refers to the ability to access, manage, understand, integrate, communicate, evaluate and create information safely and appropriately through digital technologies for different purposes.

Sustainable Development : It is the kind of development in which needs of the present generation are met without compromising on the needs of the future generations.

Inclusive Development : It refers to the development of all irrespective of gender, caste, class, religion or race.

Big Data

: Big data is the large datasets which is being generated with the help of ICTs and can be used or analysed to reach large number of people with specific aim

13.9 FURTHER READINGS

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13.10 CHECK YOUR PROGRESS: POSSIBLE ANSWER

Check Your Progress: 1

1. ICT4D refers to use of Information, Communication and Technology (ICT) for Development. This term became popular with the increasing use of ICTs in different sectors for development all over the world.

Check Your Progress: 2

1. Following are the key components of ICT4D:-
 - i. Establishment of Digital Infrastructure
 - ii. Building of Human Capacity

Check Your Progress: 3

1. By creating a gap between the people who have access to ICTs and those who do have access, digital divide has given rise to another divide between 'haves' and 'have-nots'. It has created a section of people who are not able to reap the benefits of ICT4D as they neither have the access to digital infrastructure nor have the digital skills to access the technologies.

Check Your Progress: 4

1. Participatory approaches to development are being encouraged to ensure inclusivity in ICT4D so that community members themselves can participate in each and every step of development from decision-making to generating resources for the project to actually executing the project on the ground-level and then also evaluating the project. Similarly, attention is being paid to horizontal approaches instead of solely relying on vertical or top-down approach.



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